

An adaptive intrusion detection system for the internet of things using large language models and post-quantum-secure blockchain

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ABSTRACT

The proliferation of Internet of Things (IoT) devices makes them critical infrastructure, yet their scale and openness expose them to increasing cyberattacks. Existing dynamic IoT Intrusion Detection Systems (IDSs) can partially prevent cyberattacks. However, they typically demand extensive, high-quality labeled data, which makes updates costly and labor-intensive. Recently proposed Large-Language-Model (LLM)-based IoT IDSs address data labeling. However, they often over-rely on end-to-end LLM processing instead of hybrid LLM-Machine Learning (ML) approaches, which results in a significant detection delay. Meanwhile, the IDS's data security is seldom considered in these works. This paper introduces the LLM and Post-Quantum-Secure Blockchain (QBC)-based adaptive Intrusion Detection System (LQB-IDS). LQB-IDS applies an adaptive detection method. It utilizes a lightweight ML model to detect known attacks that previously appeared in the training dataset. Meanwhile, all unknown attacks are handled by the LLM for dynamic analysis, using external knowledge secured by LLM managed QBC. Thus, it can continuously detect and adapt for unknown attacks without relying on pre-labeled data. Its core contributions include 1) an LLM-based unknown IoT traffic analyzer eliminating manual labeling; 2) a dual-switch learning approach combining LLM with lightweight ML models for low detection delay; and 3) a novel credit scoring algorithm with an LLM-Maintained Post-Quantum-Secure Blockchain-Based Database (LLM-QBC-DB) ensuring data security. Experimental results demonstrate that LQB-IDS achieves a 0.98 detection rate and a 0.97 average F1-score without labeled data, presenting an efficient and scalable solution for emerging IoT threats.

1. Introduction

The Internet of Things (IoT) spans from simple household appliances to sophisticated industrial sensors. It has quickly become a basic communication tool, changing how everyday objects interact within a linked ecosystem. IoT supports these devices to be seamlessly integrated and data exchanged by including microcontrollers, transceivers, and defined protocol stacks. This connectivity drives ideas in many different fields, greatly influencing daily life as well as industrial operations [1].

However, many IoT devices have natural characteristics that make them easy targets for today's cyber threats. These include limited resources, diverse communication methods, and often weak security features [2]. Conventional security solutions provide a fundamental layer of protection. These include firewalls, access control systems, encryption, and even blockchain techniques. However, they frequently prove inadequate on their own against dynamic attack methods [3]. Among

these solutions, Intrusion Detection Systems (IDS) have emerged as a crucial and effective security component [4]. An IoT IDS is a technology centered on traffic analysis to identify and respond to malicious activities. It monitors system activities or network traffic for malicious patterns or anomalies. This provides the essential capability to identify and alert to potential intrusions threatening IoT security [5]. Generally, IoT IDS can be categorized into Network-based Intrusion Detection Systems (NIDS) and Host-based Intrusion Detection Systems (HIDS). The scope of this paper focuses on NIDS. Fig. 1 shows the general architecture of NIDS.

Responding to these growing threats, researchers have proposed more and more advanced IoT IDSs, progressively adding modern technologies to increase their capabilities, such as Artificial Intelligence (AI). For a comprehensive overview of the field, Babulak (2023) presents a collection of works that illuminate the most effective and practical applications of AI in securing critical cyber infrastructure, highlighting

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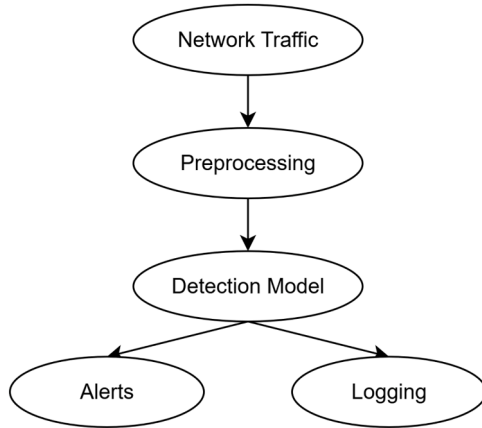


Fig. 1. General architecture of NIDS.

its indispensable role in preventing and protecting against modern cyber attacks [6]. Among AI models, Large Language Models (LLMs) have exploded in capability and application recently. They are now being investigated to improve IoT IDSs due to their sophisticated understanding of complex patterns, natural language processing capabilities for threat analysis, and their potential to automate analytical tasks [7]. This is particularly evident in their capacity to describe the intent of intrusions and abnormal behaviors [8]. This empowers traditional IoT IDSs with the novel capability to automatically acquire, organize, and assimilate external, unstructured knowledge about attacks, enabling them to achieve adaptive detection of unknown attacks without human intervention. At the same time, post-quantum secure blockchain technology [9] is emerging as another innovative way to strengthen data security and integrity in IoT ecosystems. It provides improved protection against tampering and quantum attacks [10]. However, naturally combining the various advantages of such strong but different technologies remains a challenge. Indeed, the integration of artificial intelligence, blockchain, and quantum cryptography to secure IoT ecosystems is an active and complex field of research [11].

To address this integration challenge, this paper proposes a novel LLM and Post-Quantum-Secure Blockchain-based Intrusion Detection System (LQB-IDS). The proposed LQB-IDS adopts a novel hybrid architecture that synergizes distributed and centralized methodologies. Specifically, lightweight models based on DL and ML are deployed directly on resource-constrained IoT devices for initial traffic inspection. Conversely, all LLM-based agents, such as unknown attack analyzer agents or post-quantum secure blockchain authority agents, are deployed on trusted, authoritative cloud servers, which possess substantially more computational resources for handling complex analytical tasks and system-wide coordination. The proposed system is particularly well-suited for securing critical IoT infrastructures, including smart healthcare, smart grids, and Vehicular Ad-hoc Networks (VANETs).

The rest of this paper is arranged as follows: Section 2 introduces the related work along with the contributions and novelty of this paper. Section 3 presents the system model for LLM-Maintained Post-Quantum-Secure Blockchain-Based Database (LLM-QBc-DB). Section 4 presents the details of the proposed method: LLM and Post-Quantum-Secure Blockchain-based Intrusion Detection System (LQB-IDS). Section 5 provides the numerical experiments' design, results, and the discussion for these results. Finally, Section 6 summarizes the paper.

2. Related work

This section provides a comprehensive literature review on IDSs in the IoT environment, which are essential for safeguarding against increasing security threats. IoT IDSs can be broadly classified into four categories: static, dynamic, large language model (LLM)-based, and

blockchain-based approaches. Each category offers unique mechanisms and benefits but also presents specific limitations. These methods are critically reviewed, and their key challenges are identified. Finally, the main contributions of this paper are highlighted, aiming to address some of these limitations.

2.1. Static IoT IDS

Despite the long history of static IoT IDSs, there has been a lot of creative work in this area recently. For instance, Umer Zukaib et al. [12] introduced a Meta-IDS designed for Internet of Medical Things (IoMT) networks, which employs mean-shift clustering and two-stage meta-learning with XGBoost to detect both known and zero-day attacks in real-time while ensuring data privacy. The Meta-IDS demonstrated excellent accuracy, over 99%, on the IoT Intrusion Dataset 2020 (IoTID20) dataset; however, the system has significant training resource requirements and is sensitive to network fluctuations.

Another recent study is the Multi-Layer Multi-Classification System (MLMCS) for fog-based IoT environments developed by Hossein Khosravifar et al. [13]. This system uses Dempster-Shafer theory to combine information and Group Method of Data Handling (GMDH) neural networks in layered classifiers for different feature groups. Even though this method can take a long time to train and is quite complex, it aims to be more resistant to attacks and improve the accuracy of detecting different types of attacks, achieving over 99% accuracy on Telemetry, Operating Systems, and the Internet of Things (ToN-IoT) dataset.

Apart from the previously discussed signature-based and ensemble learning approaches, reinforcement learning (RL) has also become a popular technique for IoT IDS design recently. For example, Shoujian Yu et al. suggested a new deep reinforcement learning (DRL)-based framework that used double deep Q networks with lazy penalties (DDQN-LP) with stochastic games to detect intrusions in the Industrial Internet of Things (IIoT) [14]. This algorithm automatically finds a good balance between how accurately it detects problems and how efficiently it runs by using Double Deep Q-Networks (DDQN) along with a "lazy penalty" system. Although its current model is limited in complexity and detection delay, the approach showed superior detection rates and lower overhead compared to other RL algorithms (e.g., reducing attacker success probability to ~ 0.1).

Another RL-based IoT IDS is Dual Replay Memory Reinforcement Learning (DRM-RL), which is an offline RL system developed by Ankit Sharma et al. to resolve the problem of class imbalance in network intrusion detection [15]. It prioritizes minority attack samples in a unique way by using a Dual Replay Memory (DRM) mechanism with its Deep Q-Network (DQN) agent. Compared to the usual DQN, this approach significantly improved the detection of less common attacks, such as the User-to-Root (U2R) attack by +150% and the flooding attack by +60.36% on the Aegean Wi-Fi Intrusion Dataset (AWID). The memory sampling ratio requires dataset-specific optimization, and minority class detection rates, despite improvements, might still fall short of detection rates for the majority class.

Similarly, other research has focused on developing hybrid frameworks combining machine learning and neural networks to achieve high accuracy in detecting prevalent IoT threats such as DDoS attacks on various standard datasets [16].

Even though static IoT IDSs have done well against known threats, their fixed design makes it hard for them to automatically adjust to new threats, like zero-day attacks and unfamiliar threats, as well as changes in dynamic network environments.

2.2. Dynamic IoT IDS

Recently, an increasing number of dynamic IoT IDSs have been proposed to address the limitations of static IoT IDSs discussed in the previous section. Unlike static approaches that primarily focus on improving detection performance on static datasets, dynamic IoT IDSs emphasize

adaptive capabilities for continuous detection in complex environments characterized by concept drift and unknown attacks.

Early approaches in this domain explored various pattern-based methods. For instance, Nia et al. proposed a novel approach for detecting unidentified network traffic using a pattern-based random walk [17]. Their system dynamically adapted traffic patterns in real-time and analyzed them against a threat database to identify malicious traffic. While this method showed a promising detection rate without relying on traditional machine learning models, its adaptability was dependent on the comprehensiveness of its threat database, a challenge that later dynamic IDSs sought to address with more advanced learning techniques.

One recent advanced IoT IDS is the Incremental Contrast Learning-based Intrusion Detection Framework (CAEAID), created by Zinuo Yin et al. to handle complex data and changes in patterns in IoT IDS [18]. Its main parts include a Contrastive Autoencoder (CAE) to effectively represent features and an improved Extreme Value Theory (EVT) to set flexible thresholds, along with a pseudo-labeling method that relies on CAE-EVT consistency for ongoing learning to deal with changes in data patterns. Utilizing the IoTID20 and the Canadian Institute for Cybersecurity Intrusion Detection System 2018 (CIC-IDS-2018) datasets, CAEAID demonstrated superior performance, achieving F1-scores of 91.36 % on IoTID20 and 91.30 % on CIC-IDS-2018, along with significant accuracy improvements and false positive rate reductions compared to other advanced methods. Limitations include enhancing interpretability and dynamic adaptiveness and further validation of its effectiveness in ultra-large-scale IoT deployments against highly complex real-world attack patterns.

Besides purely incremental learning-based IoT IDSs, another recent research trend in dynamic IoT IDSs is combining incremental learning and Federated Learning (FL); for example, Zhigang Jin et al. proposed the Federated Learning-based Incremental Intrusion Detection System (FL-IIDS) to address catastrophic forgetting in FL-based IDSs for dynamic environments [19]. FL-IIDS employs local clients with a dynamic example memory and a novel loss function combining Class Gradient Balance (LGB) and Sample Label Smoothing (LLS) to mitigate local forgetting, while relay clients use sample reconstruction to select the best old global model, preventing global forgetting. FL-IIDS significantly improved memory for old classes without hampering new class detection, outperforming standard approaches in incremental learning scenarios. Limitations include its primary focus on independent and identically distributed (i.i.d.) data assumptions among clients, the need to carefully determine optimal dynamic memory size, and a potential slight impact of LLS on convergence speed.

Another similar work is the Hierarchical Federated Class-Incremental Learning Network Intrusion Detection (HFIN) system, proposed by Jingxin Mao et al. to address changing network threats in IIoT environments using a layered cloud-edge-client setup [20]. To accommodate new and old threat classes, HFIN uses Class-Incremental Learning (CIL) with exemplar memory on edge servers and a Weighted Transmission Optimization (WTO) approach to prioritize critical data transmission from clients, balancing detection with resource constraints. Ranked on the NetFlow University of Queensland Network Intrusion Detection System version 2 (NF-UQ-NIDS-v2) and NetFlow The University of New South Wales' IoT Network version 2 (NF-ToN-IoT-v2), HFIN reportedly outperformed baselines by about 10 % in accuracy and F1 score. However, the need for large-scale, manually labeled data to support its incremental learning capabilities limits this system.

Further advancing this direction, another recent study proposed an integrated federated learning framework specifically optimized for Industrial IoT (IIoT) environments [21]. Their approach combines Convolutional Neural Networks (CNNs) with federated learning and employs differential privacy mechanisms to enhance data security. The work highlights how such frameworks can serve as a robust and scalable alternative to traditional centralized machine learning methods.

However, while these federated learning approaches, including FL-IIDS, HFIN, and the aforementioned frameworks, effectively address data privacy concerns by keeping raw data decentralized, they share common vulnerabilities. A primary concern is their susceptibility to data and model poisoning attacks, where malicious nodes can intentionally submit corrupted model updates to degrade the global model's performance. Furthermore, their capacity for detecting novel zero-day attacks is inherently constrained by the diversity of data available across the distributed client nodes. This dependency can hinder the system's adaptability when confronted with entirely new threat vectors not previously encountered by any participating devices.

Apart from combining incremental learning and federated learning, other recent dynamic IoT IDSs focus on particular applications; for example, Qiuyu Lu et al. proposed AdaOIL-IDS, an adaptive online incremental learning system for Network Intrusion Detection Systems (NIDS) in Smart Grids (SGs), to address constantly evolving cyberattacks [22]. Along with an adaptive learning framework with a dynamic evaluator to alternate between online learning and offline retraining for lifelong learning, AdaOIL-IDS boasts a unique Class-Correlated Broad Learning System (CC-BLS) using a weighted Class-Correlated factor (CC-factor) for improved accuracy and efficient incremental updates. Tests show that CC-BLS starts with a high accuracy of 95.70 % and maintains good performance at 90.91 % after 12 rounds of online learning, performing better than models like Broad Learning System-Class Incremental Learning (BLS-CIL), Incremental Classifier and Representation Learning (iCaRL), and Elastic Weight Consolidation (EWC). The limitations of this work include some accuracy reduction following online learning compared to stationary models, increased update times for CC-BLS due to CC-factor calculations, and an inherent trade-off between maximal accuracy and the computational efficiency prioritized by its adaptive framework.

Other examples include our previous research on the Incremental Lifetime Learning Intrusion Detection System (ILL-IDS) [23] and the Automatic Incremental Lifetime Learning Intrusion Detection System (AILL-IDS) [24], designed for a Vehicular Ad-hoc Network (VANET). ILL-IDS applies incremental Support Vector Machine (SVM) techniques, while AILL-IDS applies a density-based stream model (DBSTREAM) to dynamically detect unknown attacks in such a network. Both of them can achieve a detection rate of more than 0.97 on a hybrid dataset formed by the Vehicular Reference Misbehavior (VeReMi) and the VeReMi Extension datasets. Compared to ILL-IDS and other dynamic IoT IDSs, AILL-IDS utilizes semi-supervised learning to reduce its reliance on labeled data for model updating, but it still depends on such data. Some other recent dynamic or adaptive IoT IDS works include: Halgamuge and Niyato [25] and Chen et al. [26].

Although dynamic IoT IDSs possess lifelong learning capabilities that enable adaptive detection of unknown attacks and effective handling of concept drift, a major challenge lies in their reliance on a continuous supply of high-quality, manually labeled data for model updates throughout the entire learning process. Furthermore, other advanced deep learning models, such as the cross-domain IDS using a BiGRU-LSTM-Attention mechanism, have also been proposed to enhance security for medical and industrial IoT by leveraging robust feature extraction and attention-based classification, demonstrating another direction in designing sophisticated dynamic detection frameworks [27]. In parallel, Transformer-based architectures have shown significant promise for classifying multivariate time-series data, which is analogous to network traffic analysis. For instance, the Knowledge Aggregation Transformer Network (KATN) effectively combines local feature extraction with the global relational awareness of Transformers to achieve state-of-the-art performance in this domain [28]. While such powerful models offer high accuracy, their computational complexity often makes them less suitable for real-time inference on resource-constrained IoT edge devices. Similarly, while advanced knowledge distillation techniques can facilitate model updates by transferring knowledge between heterogeneous

models [29], they traditionally rely on a pre-existing, accurately trained “teacher” model. In conclusion, the dependence on labeled data, either directly for training or for creating a teacher model, presents a significant barrier to the widespread deployment and practical application of such systems in real-world environments.

2.3. LLM-based IoT IDS

To reduce the heavy reliance on high-quality labeled data in dynamic IoT IDSs employing traditional Machine Learning (ML) or Deep Learning (DL) models for incremental learning or retraining, recent advances in Large Language Models (LLMs) have inspired researchers to leverage their powerful analytical capabilities for dynamic analysis of environmental changes and the detection of unknown attacks, leading to the emergence of LLM-based IoT IDSs [7].

A common approach to employing LLMs in IoT IDS detection is to directly utilize them for analyzing IoT traffic. For instance, Alaeddine Diaf et al. proposed Bidirectional and Auto-Regressive Transformers Predict (BARTPredict), a proactive intrusion prediction framework [27]. It employs fine-tuned Bidirectional and Auto-Regressive Transformers (BART) for packet prediction and classification, alongside fine-tuned Bidirectional Encoder Representations from Transformers (BERT) for assessment, all operating at the Multi-Access Edge Computing (MEC) level and innovatively using LLMs to anticipate attacks by analyzing packet headers, which enhances privacy and efficiency. While achieving 98% overall detection accuracy on the Canadian Institute for Cybersecurity Internet of Things Dataset 2023 (CIC-IoT-2023), the framework requires further refinement, model optimization, and more extensive validation, particularly due to limited data during fine-tuning and the need to assess generalization across diverse scenarios. Along this line of research, other notable frameworks include SecurityBERT, which introduces a privacy-preserving lightweight model based on BERT for IoT devices [30], and HuntGPT, which integrates machine learning-based anomaly detection with LLMs to provide explainable AI capabilities [31].

Beyond directly applying LLMs to analyze IoT traffic, another integration approach involves using LLMs to interpret the outputs of traditional machine learning (ML) or deep learning (DL) models, which perform the initial traffic analysis. For instance, Yanjie Li et al. proposed IDS-Agent [32]. IDS-Agent is presented as the first AI agent powered by LLMs for IoT intrusion detection, using an LLM to orchestrate various ML classifiers (e.g., Random Forest (RF) and K-Nearest Neighbors (KNN)), knowledge retrieval via Retrieval-Augmented Generation (RAG), and Long-Term Memory (LTM) in a reasoning-action pipeline to make and explain decisions. It uniquely provides explainable results and customizable sensitivity, demonstrating strong zero-day attack detection (0.61 recall on unseen attacks from the CIC-IoT-2023) and outperforming state-of-the-art baselines on benchmarks like the Army Cyber Institute Internet of Things Network Traffic Dataset 2023 (ACI-IoT-2023) and CIC-IoT-2023 (F1-scores 0.97 and 0.75, respectively). A notable trade-off is the increased processing time (e.g., 8.65 s per instance) resulting from its comprehensive knowledge retrieval and aggregation processes, especially when compared to simpler LLM-based approaches; however, this delay is deemed acceptable for real-time applications.

Another way to leverage LLMs in IoT IDSs is by integrating them into rule-based systems, where LLMs can automatically generate and dynamically update detection rules. A recent work in this area, proposed by Xiaowei Hu et al., introduces an LLM-based agent [33]. This agent uses an LLM within a RAG framework guided by Chain of Thought (CoT) prompting from processed vulnerability reports and malicious payloads to create and generalize IDS rules. This system features a distinct automatic rule generalization scheme that broadens detection for attack variants, achieving a high accuracy of 0.996, perfect precision of 0.99, and a zero false positive rate when generating rules from diverse inputs. A key limitation is the potential for LLM hallucination to produce malformed or erroneous rules, necessitating a dedicated inspection and

post-processing validation step to ensure rule validity and correct formatting for systems like Suricata or Snort.

In summary, although LLM-based IoT IDSs demonstrate strong capabilities in automated attack analysis and reduce the dependence on high-quality labeled data, most of their efforts have been on improving the LLM’s analysis skills, like using RAG or better prompt engineering [34], without much focus on the security of the LLMs themselves; as a result, issues like data poisoning and other attacks on LLMs could seriously weaken the effectiveness of these advanced IDS solutions.

2.4. Blockchain-based and post-quantum-secure blockchain-based IoT IDS

To tackle the growing concern of attacks on the core models behind IoT intrusion detection systems, such as data poisoning or tampering during training updates, researchers are turning to more secure solutions. One promising direction is the integration of technologies like blockchain or even post-quantum-secure blockchain, which offer built-in protection through encryption and tamper resistance. By combining these technologies with IDSs, it is possible to strengthen their defenses and improve overall trust in the system’s reliability and resilience.

One recent blockchain-based IoT IDS is the Blockchain-based Intrusion Detection System (BIDS) for the Internet of Vehicles (IoV), proposed by Meisam Sharifi Sani et al. [35]. BIDS tracks how vehicles move using a blockchain, where each block is verified when a vehicle reaches a specific place and time; if there are any mistakes, the following blocks become invalid, preventing wrong information from affecting the Traffic Management System (TMS). This decentralized system employs a permissioned blockchain, wherein Road Side Units (RSUs) authenticate vehicle locations, and the Elliptic Curve Digital Signature Algorithm (ECDSA) ensures secure communications, facilitating the transparent identification of malicious mobility patterns without dependence on a central authority. In tests using the Opportunistic Network Environment (ONE) simulator, BIDS was able to detect issues with high accuracy (up to 98%) and outperformed methods like ART and TMEC in terms of precision, recall, and F-measure when faced with Man-in-the-Middle (MiTM) attacks. Key limitations include the trade-off between RSU density versus cost and detection time and high overhead from frequent vehicle mobility updates impacting cloudlet mining and RSU communication.

However, traditional blockchain may be vulnerable to quantum computing-based attacks in the near future; therefore, some researchers are also attempting to use post-quantum secure blockchain to construct IoT IDSs to prevent quantum attacks. For instance, Mohammad Tazeem Naz et al. proposed the Self-Defensive Post-Quantum Blockchain Architecture (SD-PQBA) [36], designed to secure Supervisory Control and Data Acquisition (SCADA) systems in smart grids against both classical and quantum cyber threats. This system combines a blockchain strengthened with Post-Quantum Cryptography (PQC) (using a 3-key method, possibly involving lattice-based techniques and Quantum Key Distribution (QKD) for sharing keys) and introduces a new Proof of Derived Authority (PoDA) agreement method designed for SCADA systems. While SD-PQBA aims for real-time efficiency and low overhead suitable for IoT nodes, its theoretical nature requires empirical validation, and it faces challenges including the computational demands of PQC on resource-constrained devices, potential latency, scalability issues for vast SCADA data, and significant infrastructure changes for implementation.

While blockchain and post-quantum secure blockchain offer robust protection for the IoT IDS itself, there is a notable scarcity of designs employing LLMs to manage post-quantum secure blockchains; thus, effectively integrating the synergistic advantages of LLMs and post-quantum secure blockchain to develop efficient, truly dynamic IoT IDSs remains a key research gap.

2.5. Limitations of existing IoT IDS approaches

Based on the above discussions, the primary limitations of existing IoT IDSs can be summarized as follows:

- **Static IoT IDSs:** Their fixed design inherently hinders adaptation to novel threats and concept drift.
- **Dynamic IoT IDSs:** They often require substantial high-quality, manually labeled data for effective lifelong learning and model updates, limiting practical deployment.
- **LLM-Based IoT IDSs:** Current research predominantly focuses on enhancing analytical capabilities, often misconsidering the security vulnerabilities of the LLMs themselves from data tampering, such as data poisoning.
- **Blockchain-Based and Post-Quantum-Secure Blockchain-Based IoT IDSs:** While improving data integrity and security (with quantum versions addressing quantum threats), they do not inherently solve adaptive detection challenges and often lack integration with advanced dynamic management systems like LLMs.

2.6. Contributions and novelties

To address the limitations discussed above, LQB-IDS is introduced in this paper. The proposed LQB-IDS offers the following key contributions and novel features:

Distinct from purely distributed frameworks like federated learning which prioritize data privacy through decentralized training, our proposed LQB-IDS introduces a novel hybrid architecture. This architecture strategically combines the strengths of both centralized and distributed models. Specifically, it delegates high-speed, real-time detection of known threats to lightweight models at the network edge, functioning in a distributed manner. Meanwhile, the computationally intensive tasks such as the dynamic analysis of unknown attacks and the governance of system-wide trust are handled by a powerful, centralized LLM on a secure cloud server. The secure and trustworthy communication between the centralized intelligence and the distributed edge nodes is facilitated by the LLM-Maintained Post-Quantum-Secure Blockchain-Based Database (LLM-QBc-DB). This hybrid approach not only preserves the low-latency benefits of edge detection but also overcomes critical limitations of purely decentralized systems, namely their vulnerability to data poisoning attacks and their inherent difficulty in adapting to novel threats without a centralized knowledge nexus.

1. Proposing an LLM-based framework capable of dynamically analyzing unknown attacks by autonomously gathering and leveraging external knowledge, thereby eliminating the need for manual data labeling.
2. Proposing a dual-switch learning method that distills knowledge from a resource-intensive, cloud-based LLM to a lightweight edge ML model. This method dynamically switches between edge ML model operational modes (training/prediction) and output sources (LLM/ML model), which significantly reduces detection latency and accelerates performance recovery from concept drift, surpassing state-of-the-art dynamic IDSs in adaptation speed.
3. Proposing a novel credit score calculation algorithm integrated with a post-quantum secure blockchain governed by an LLM. This synergy ensures the security, integrity, and non-repudiation of shared IDS data from cloud to edge IoT devices, providing a robust defense against data poisoning, collusion, and emerging quantum threats to the system's infrastructure.

To facilitate a clearer comparison of our proposed LQB-IDS with other IoT IDS approaches, Table 1 summarizes the key designs and features of existing IoT IDSs reviewed in Sections 2.1 to 2.4, alongside those of LQB-IDS. A detailed explanation of our proposed system will be provided in the subsequent sections of this paper.

Table 1

Summary of existing IoT IDS studies and proposed study.

IDS Name	Dynamic Adaptation	Automated Labeling Capability	Data Tampering Resilience	Post-Quantum Security
Meta-IDS [12]	×	×	×	×
MLMCS [13]	×	×	×	×
DDQN-LP [14]	×	×	×	×
DRM-RL [15]	×	×	×	×
CAEAID [18]	✓	×	×	×
FL-IDS [19]	✓	×	×	×
HFIN [20]	✓	×	×	×
AdaOIL-IDS [22]	✓	×	×	×
ILL-IDS [23]	✓	×	×	×
AILL-IDS [24]	✓	×	×	×
HuntGPT [31]	✓	✓	×	×
SecurityBERT [30]	✓	×	×	×
BiGRU-LSTM-Attention [27]	✓	×	×	×
KATN [28]	×	×	×	×
HMKD [29]	×	×	×	×
BARTPredict [37]	✓	×	×	×
IDS-Agent [32]	✓	✓	×	×
LLM-based agent (rule gen.) [33]	✓	×	×	×
BIDS [35]	×	×	✓	×
SD-PQBA [36]	×	×	✓	✓
LQB-IDS (Ours)	✓	✓	✓	✓

3. System model for LLM-maintained post-quantum-secure blockchain-based database (LLM-QBc-DB)

3.1. Post-quantum-secure blockchain (QBc)

While secure against classical computers, traditional blockchain systems face a significant future threat from the advancement of quantum computing. Cryptographic algorithms like RSA, commonly used for the digital signatures that ensure data integrity, are theoretically vulnerable to attacks from quantum computers running algorithms such as Shor's. To counter this emerging threat, this paper adopts the principles of a Post-Quantum-Secure Blockchain (QBc). A QBc enhances a standard blockchain by replacing quantum-vulnerable cryptographic components with algorithms resistant to quantum attacks, a field known as Post-Quantum Cryptography (PQC).

Among the various PQC approaches, Lattice-based Cryptography is a leading candidate, as its underlying mathematical problems are considered computationally infeasible to solve in a practical timeframe, even for quantum computers. The PQC features in this work are based on the principles of Dilithium, a prominent digital signature scheme built upon lattice-based cryptography. By integrating a PQC signature scheme like Dilithium for critical operations, such as digitally signing and verifying transactions on the ledger, the blockchain can ensure the long-term security, integrity, and authenticity of its stored data against threats from both classical and future quantum adversaries.

3.2. Structure of LLM-QBc-DB

The integrity of data is important for an IoT IDS, as these systems are often targeted by data tampering-based attacks like adversarial attacks [38] and data poisoning [39]. An LLM-maintained post-quantum secure blockchain-based database (hereinafter referred to as LLM-QBc-DB) is proposed to counter such threats and support the dynamic adaptation of LQB-IDS. LLM-QBc-DB is crucial for storing data to support adaptive model updating for LQB-IDS, including unknown attack samples, structured knowledge on emerging threats, the output of LLM-based unknown IoT traffic analysis module and updated weights for edge models. LLM-QBc-DB has two major components: a post-quantum secure blockchain (QBc) and an LLM-based Post-Quantum-Secure Blockchain Authority Agent (LLM-QBc-AA).

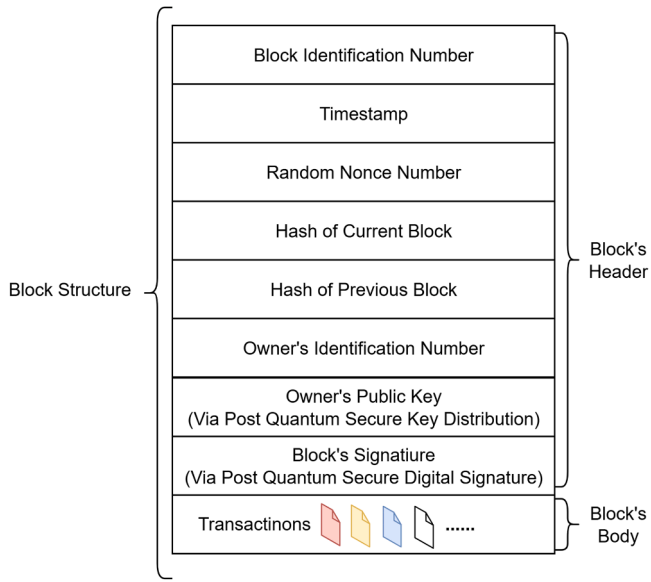


Fig. 2. Block structure of QBC.

The post-quantum secure blockchain component leverages the inherent cryptographic, distributed storage, and non-repudiation characteristics of blockchain technology, offering a robust defense against data tampering-based attacks. Recognizing the potential threat posed by advancements in quantum computing to conventional blockchain encryption [40,41], LQB-IDS applies a post-quantum secure blockchain (QBC), which has a similar architecture to conventional blockchains but is enhanced with post-quantum secure algorithms, such as post-quantum secure key distribution [42] and lattice-based cryptography [43], to ensure resilience against quantum attacks. Fig. 2 shows block structure of the QBC applied.

Due to the computational limitations of IoT devices, a Proof-of-Authority (PoA)-based consortium blockchain framework is adopted for the post-quantum secure blockchain, as it is more lightweight compared to Proof-of-Work (PoW)-based public blockchains. However, the core challenge of any PoA system lies in the governance of its authority nodes. Traditionally, this requires manual oversight to detect and remove malicious entities, which is impractical and costly in a large-scale, dynamic IoT environment. To address this gap, the LLM-based Post-Quantum-Secure Blockchain Authority Agent (LLM-QBc-AA), an autonomous governance mechanism, is introduced. This agent leverages the advanced reasoning capabilities of an LLM to interpret complex behavioral data, automating the traditionally manual process of ensuring node trustworthiness.

The Post-Quantum-Secure Blockchain (QBC) in LQB-IDS operates with three distinct types of nodes, each assigned specific roles and responsibilities concerning block creation and message handling:

- **Authority Node:** These nodes are responsible for blocking the corrupted nodes and the formation and broadcasting of all new blocks within the QBC.
- **Trusted Node:** When an authority node receives messages from a trusted node, it directly incorporates these messages into a new block without requiring further validation of the source's credibility.
- **Ordinary Node:** For messages originating from an ordinary node, an authority node first employs a Credit Score Calculation Algorithm (which is detailed in a later section) to assess the message's credibility. Only after a satisfactory credit score is confirmed will the authority node include the message content in a new block.

Table 2 summarizes the different node types within the QBC in LQB-IDS and their corresponding system modules. Fig. 3 shows the structure of LLM-QBc-DB applied in LQB-IDS.

Table 2

Node types and corresponding module (s) in the LLM-QBc-DB.

Node Type	Corresponding LQB-IDS Module(s)
Authority Node	LLM-Based QBc Authority Agent (LLM-QBc-AA)
Trusted Node	LLM-Based Summarizer
	LLM-Based Analyzer
Ordinary Node	IoT Edge Detector
	Dual-Switch Mechanism-based Adaptive Model Updater
	Open-source IoT attack knowledge resources

4. LLM and post-quantum-secure blockchain-based intrusion detection system (LQB-IDS)

4.1. Motivation

Existing IoT IDSs generally suffer from two critical limitations that motivate this research. Firstly, there is a discernible gap between dynamic analytical capabilities and system self-protection. LLM-based IoT IDSs for dynamic attack analysis often ignore robust mechanisms to protect the IDS itself, especially the LLM applied in the IDS. Conversely, IoT IDSs that implement strong self-protection mechanisms, such as blockchain or post-quantum secure blockchain technologies, usually depend on simpler, static IDS models and do not possess efficient adaptive detection capabilities. As best as our understanding, no current work effectively integrates the advanced dynamic and adaptive attack analysis offered by LLMs with the inherent security and data integrity advantages of post-quantum secure blockchain.

Secondly, current adaptive dynamic updating methods for IoT IDSs exhibit inherent deficiencies. These methods can be categorized into two primary approaches. The first approach applies machine learning (ML) or deep learning (DL) models for inference, along with various methods for updating these models to attain adaptive capabilities. These systems typically have low detection delay, similar to conventional ML/DL models; however, they frequently experience a notable decline in detection rates during the model update period. The second approach continues to apply LLMs to gather attack-related knowledge from diverse sources and to conduct automatic analyses of attacks. This method mitigates the problem of detection rate decline during model updates, as the core LLM model is not frequently updated. Instead, updates concentrate on the attack-related knowledge database. However, it generally has a higher detection delay compared to ML/DL-based IoT IDSs. To the best of our understanding, no existing work has successfully combined the low-latency characteristic of ML/DL models with the LLM's strength in maintaining detection performance during knowledge updates, thereby avoiding the vulnerability window of traditional model retraining.

4.2. Overview of LQB-IDS

Inspired by these limitations, this paper proposes LQB-IDS, a novel framework designed to address these challenges synergistically. LQB-IDS leverages LLM-QBc-DB to bridge the gap between dynamic analysis and system self-protection. This component uniquely combines the strengths of LLMs and post-quantum secure blockchain technology. In this architecture, the LLM is employed to automatically analyze attacks and dynamically manage the authority nodes that govern the post-quantum secure blockchain. The unchangeable and secure nature of the post-quantum secure blockchain creates a safe space for the LLM and its data, effectively protecting against harmful data attacks. Additionally, to address the shortcomings of current update methods, LQB-IDS uses a dual-switch learning mechanism based on knowledge distillation. This approach utilizes lightweight ML models as the primary model for high-speed inference under normal operational conditions. When unknown attacks appear and the primary ML models need to be updated, the system dynamically switches to the LLM-based unknown IoT traffic analysis module. The labeled data generated by the this module is

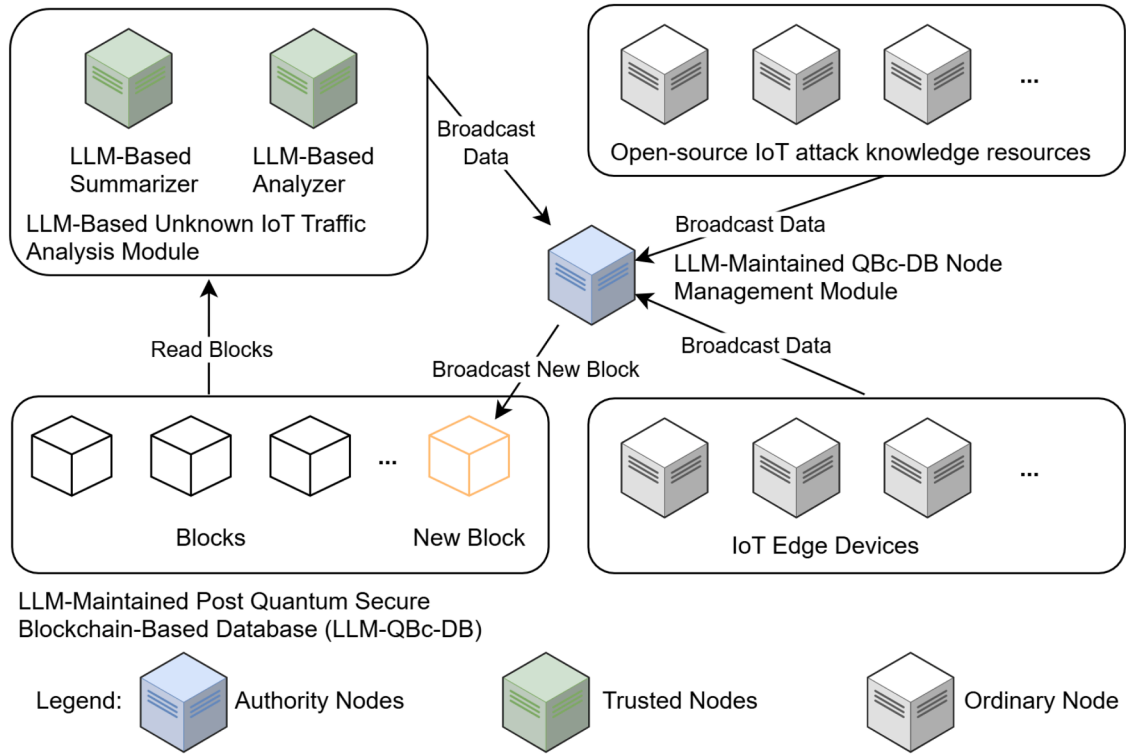


Fig. 3. Structure of LLM-QBc-DB.

then used to automatically retrain and update the primary ML model via knowledge distillation until it is successfully updated. This strategy is engineered to achieve an extremely low average detection delay and significantly shorten the vulnerable window during model updates in conventional dynamic IDSs. Through this design, LQB-IDS not only aims to achieve efficient adaptive detection but also emphasizes the clarity of its decision logic. Ensuring the algorithmic transparency of critical mechanisms, such as the model switching, is essential for establishing a trustworthy IDS in the IoT environment, a principle consistent with that emphasized in other IoT security domains like lightweight cryptography [44]. Fig. 4 shows the overall architecture of LQB-IDS.

4.3. Role of each model

The proposed LQB-IDS framework is composed of four core modules, each is designed to perform a specific function within the system's adaptive detection and security architecture. The roles of these key components are summarized as follows:

- **The IDS Edge Detector:** Acting as the system's frontline sensor, this hybrid module resides on edge devices. Its role is to perform high-speed initial analysis, rapidly identifying known attacks while isolating novel, unknown IoT traffic patterns for deeper inspection.
- **LLM-Based Unknown IoT Traffic Analysis Module:** This module serves as the system's central intelligence core. It autonomously processes the "unknown IoT traffic" flagged by the edge detector, using a multi-step analytical process to classify new threats and generate actionable labels without requiring manual intervention.
- **LLM-QBc-DB Node Management Module:** This module provides the secure and trusted backbone for the entire framework. Its role is to ensure the integrity of all shared knowledge using post-quantum cryptography, while also autonomously governing the network nodes to detect and neutralize malicious actors, thus preventing data poisoning and collusion attacks.
- **Dual-Switch Mechanism-based Adaptive Model Updater:** This module orchestrates adaptive lifelong learning. When concept drift is

detected, its dual-switch mechanism temporarily redirects traffic to the LLM-based analyzer for accurate, real-time decisions, thus avoiding a drop in detection rates. Simultaneously, it uses the LLM's output to retrain the lightweight edge model via knowledge distillation, ensuring the system adapts to new threats without compromising its low detection delay during normal operations.

4.4. Modules in the LQB-IDS

4.4.1. The IDS edge detector

The LQB-IDS edge detector utilizes a hybrid structure to ensure minimal detection delay. This structure consists of two key components: (1) a simple and efficient ML-based model for detecting known attacks and (2) a DL-based model for identifying unknown IoT traffic. Evaluations of different machine learning classifiers on the CIC-IoT-2023 dataset have indicated that algorithms like Decision Tree and Random Forest (RF) generally perform better than many other algorithms in classifying cyberattacks [45]. As a result, LQB-IDS applies an RF model as its ML-based intrusion detector. This model is applied for attack detection in a normal case when no new types of attacks or concept drift appear in the IoT network environment. Although the RF-based intrusion detector generally achieves a high detection rate under the condition the only known attack appears, it is inherently limited in its ability to handle unknown attacks, and concept drift occurs in the IoT network environment. To overcome this challenge, instances of conceptual drift and unknown attacks are categorized as "unknown IoT traffic." Then apply an unsupervised DL-based unknown IoT traffic detector to perform a binary classification, separating known IoT traffic and this newly defined unknown IoT traffic. In this paper, a Denoising Convolutional Autoencoder (DCAE) [46] is applied as the model for the DL-based unknown IoT traffic detector. More details of the DCAE model can be find in [Appendix B](#).

The operation of IDS edge detector is given below:

- Firstly, incoming IoT traffic is replicated into two identical streams.

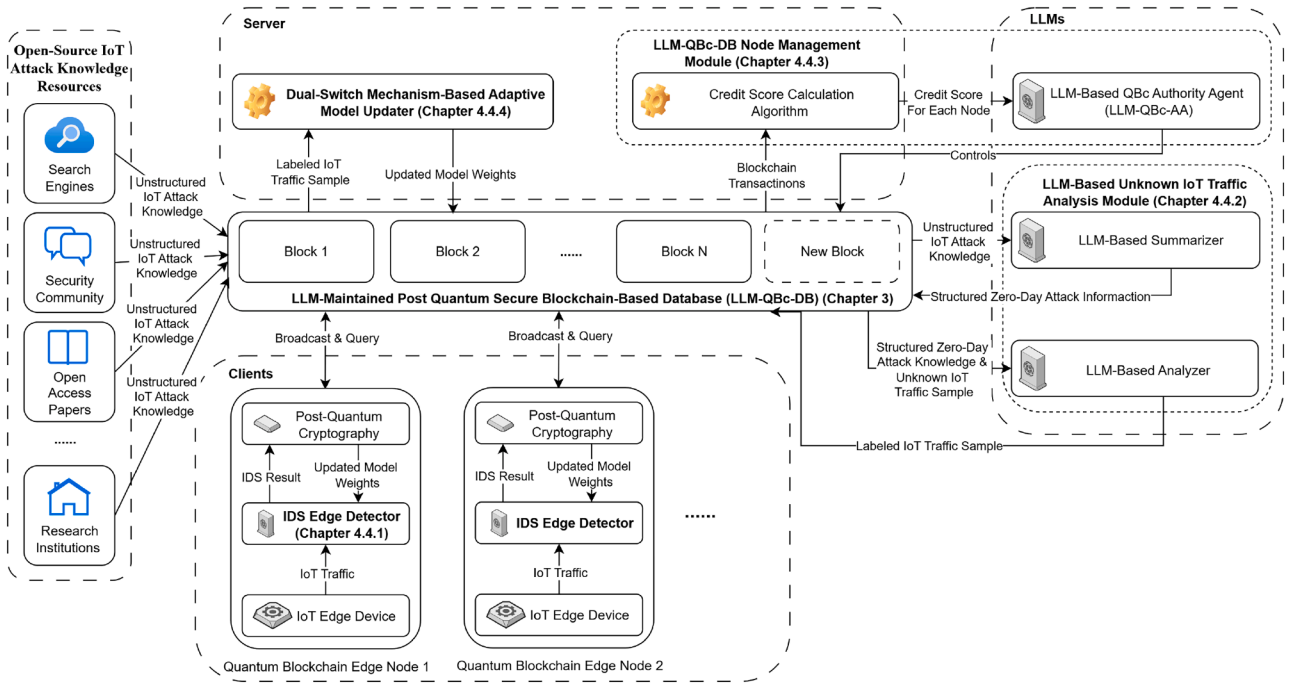


Fig. 4. Architecture of LQB-IDS.

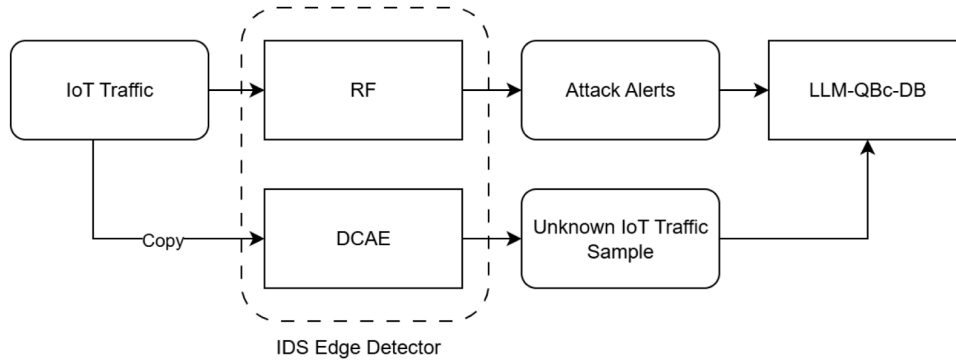


Fig. 5. Working procedure of IDS edge detector.

- ii. Then the input traffic is processed by a parallel processing method: one stream is fed to a Random Forest (RF) model to detect known attacks; The second stream is processed by the DCAE model to identify unknown traffic that differs from both the normal behavior and the known attacks learned by RF model during training.
- iii. Finally, as an ordinary node, the IDS edge detector broadcasts its detection results, including both detected attack data and unknown IoT traffic, to the post-quantum secure blockchain.

Fig. 5 shows the working procedure of the IDS edge detector.

4.4.2. LLM-based unknown IoT traffic analysis module

Once the IDS Edge Detector identifies and forwards ‘unknown IoT traffic’ to the blockchain, the LLM-based unknown IoT traffic analysis module takes center stage. This system is specifically designed to reduce the reliance on manually labeled data by autonomously analyzing these new and unclassified threats. It has two major components: an LLM-based summarizer and an LLM-based analyzer. The LLM-based summarizer is responsible for gathering unstructured attack-related knowledge from open-source IoT attack knowledge resources, such as security forums, open-access academic publications from universities, and reports from security research institutions. It then processes and transforms this information into structured, retrievable knowledge. Subsequently, the

LLM-based analyzer utilizes this structured knowledge to analyze unknown IoT traffic samples, determining their nature and categorizing them based on statistical features. This two-stage process enables the fully automatic analysis of unknown IoT traffic, thereby eliminating reliance on manually labeled data for lifelong system adaptation. Fig. 6 shows the structure of the LLM-based unknown IoT traffic analysis module.

LLM-based summarizer. In LQB-IDS, an LLM-based summarizer has been designed to function as a trusted node within the post-quantum secure blockchain. Its purpose is to organize the filtered IoT attack knowledge in the blockchain, sourced from open-source resources, such as security forums, open-access academic publications from universities, and reports from security research institutions, into JavaScript Object Notation (JSON)-formatted structured data. The LLM-based summarizer employs a specific template (see Appendix C) to guide the LLM in structuring this data. Subsequently, the LLM-processed data undergoes regular expression-based format validation before being uploaded to the LLM-QBc-DB.

LLM-based analyzer. The LLM-based analyzer plays a crucial role in categorizing unknown IoT traffic. After the DL-based unknown IoT traffic detector uploads the DCAE latent space representations of unknown IoT

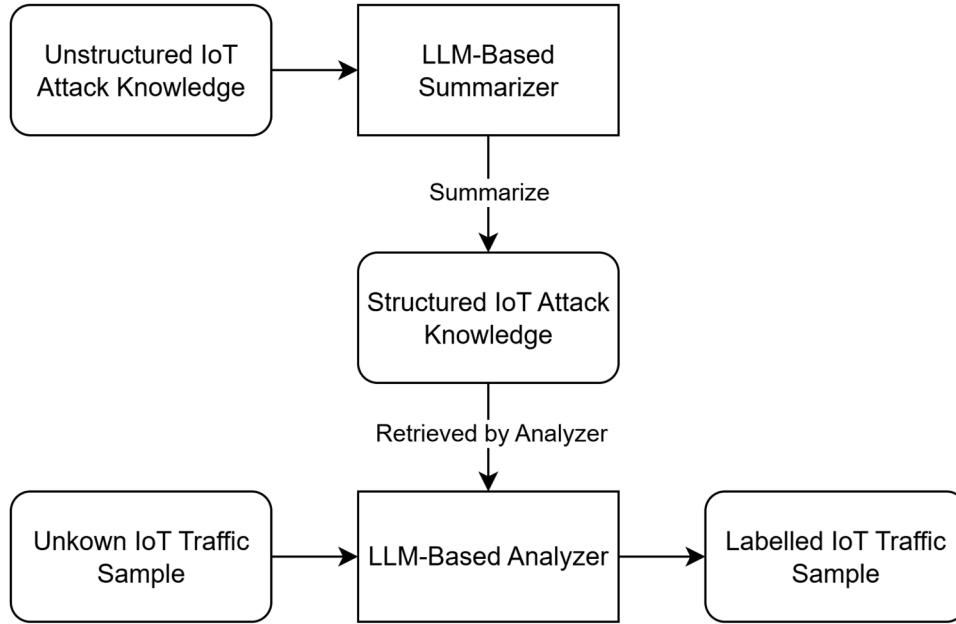


Fig. 6. Structure of LLM-based unknown IoT traffic analysis module.

traffic samples to the LLM-QBC-DB and the LLM-based summarizer contributes structured attack knowledge, the LLM-based analyzer retrieves both sets of information from the blockchain. It then leverages the collated attack knowledge to conduct a detailed analysis of the unknown IoT traffic samples to ascertain their specific attack categories or false positives caused by concept drift in the IoT network environment. This analysis is performed in three major steps:

Firstly, the unknown IoT traffic samples undergo a clustering process. This clustering is executed based on the DCAE latent space representations of these samples. After testing different clustering algorithms, such as k-means, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), or hierarchical clustering, k-means was selected due to its superior balance of accuracy and computational speed. As the precise number of attack categories (the k value) within the unknown samples is not predetermined, an iterative approach is employed. Different k values are tested, and the Silhouette Score [47] is calculated for each to compare the performance of the k-means clustering, thereby identifying the most appropriate k parameter for the algorithm.

Secondly, once the unknown IoT traffic samples have been clustered, the next step is to calculate the statistical characteristics for each feature within every cluster. These statistical features are then transformed into natural language representation using a predefined template via predefined templates, as specified in C.

Finally, the natural language descriptions generated for each cluster are utilized to perform RAG against the entire corpus of attack knowledge. The resulting retrieval-augmented prompt is then fed into the LLM applied in the LLM-based analyzer. The LLM analyzes this enriched prompt to determine the specific attack category for the samples contained within that cluster. As a trusted node, the LLM-based analyzer broadcasts its analytical findings back to the blockchain. Ensuring the integrity and security of this newly generated labeled data is paramount for the system's long-term reliability. This critical task of secure data management is handled by the LLM-Maintained QBC and its associated node management mechanisms, as will be detailed in the next section.

4.4.3. Dual-switch mechanism-based adaptive model updater

To ensure LQB-IDS can effectively respond to unknown attacks and evolving network environments, an adaptive lifelong learning framework is essential. This adaptive lifelong learning framework can be described formally as follows: Let X denote the complete set of input

data that LQB-IDS processes during its operational lifetime. The LQB-IDS model undergoes N updates. At stage t , the model is represented as Ω_t , and the data processed at this stage as the subset X_t ; here, t is an index ranging from 0 to N . Initially, the data subset X_0 is used to train the inaugural model Ω_0 . Subsequently, for each t from 0 to $N - 1$ (covering the N updates), the model version Ω_{t+1} is produced by updating the existing model Ω_t using the data subset X_t . Fig. 7 illustrates the lifelong learning process of the LQB-IDS detection model.

While traditional lifelong learning methods try to keep adapting for ongoing detection, they often face problems during updates that can lower detection rates for a while, or they can become slow and use a lot of computing power if updates happen too often [48]. To mitigate these issues, LQB-IDS introduces a novel adaptive lifelong learning framework that is centered around a dual-switch learning strategy. This strategy is engineered to achieve two primary goals: maintaining a consistently low detection delay during normal operations and minimizing the performance degradation window during model updates. At its core, the approach intelligently alternates between a primary lightweight model for rapid inference and an auxiliary LLM-based system that dynamically analyzes unknown IoT traffic. This LLM-based system does immediate analysis and facilitate the efficient retraining of the main model via knowledge distillation until the primary model is successfully updated. This dual-switch approach is designed to maintain a consistently low detection delay while minimizing the performance decline window during the model update period.

The dual-switch learning works in two distinct phases: a normal operational phase and a model update phase. The transition between system states is not arbitrary. It is governed by a KSWIN (Kolmogorov-Smirnov Windowing)-based concept drift detector [49], which triggers the shift from the normal operational phase to the model update phase by monitoring the volume of 'unknown IoT traffic' identified by the IDS Edge Detector. The node reverts to the normal operational phase once the ML-based intrusion detector has completed its update.

The KSWIN algorithm is a drift detection method that identifies changes in the underlying distribution of a data stream. Its key strength is being non-parametric, meaning it doesn't require any assumptions about the specific distribution of the data (e.g., that it is normal).

The algorithm functions by maintaining a sliding window, Ψ , of size n . From this window, it compares two subsets of data:

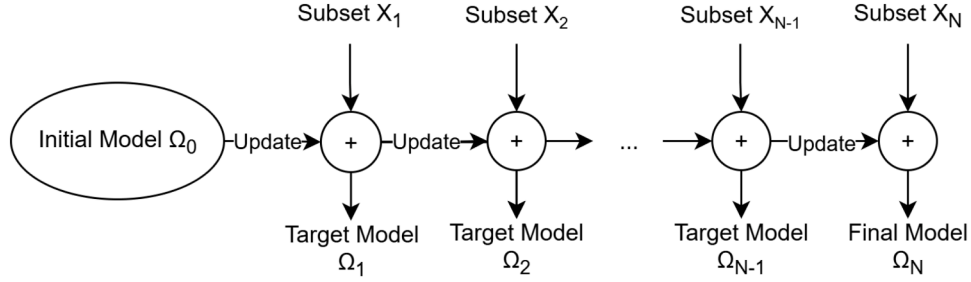


Fig. 7. Lifelong learning process of the detection model.

- **R**: A set composed of the most recent r samples, representing the current data concept.
- **W**: A set of r samples chosen randomly from the older part of the window, which approximates the past data concept.

It then applies the Kolmogorov-Smirnov (KS) statistical test to measure the distance between the empirical distributions of R and W . A concept drift is flagged if this distance surpasses a statistically determined threshold. The formal condition for detecting a drift is:

$$\text{dist}(R, W) > \sqrt{-\frac{\ln(\alpha)}{2r}} \quad (1)$$

Here, r is the size of the compared samples (R and W), and α is the significance level, which defines the acceptable probability of a false positive detection.

The knowledge distillation-based dual-switch learning strategy is governed by two switches controlled by the KSWIN algorithm, *switch_0* and *switch_1*, which manage the operational flow and output of the LLM-based unknown IoT traffic analysis module and the ML-based intrusion detector. *Switch_0* dictates whether the ML-based intrusion detector is in inference mode or training mode. *Switch_1* is responsible for routing the output from the currently active classifier to the node's final prediction output.

The operational logic for these switches during each phase is as follows:

- **Normal Operational Phase:**
 - *Switch_0* configures the ML-based intrusion detector to operate in inference mode.
 - *Switch_1* routes the output from the ML-based intrusion detector to be the node's final output.
- **Model Update Phase:**
 - *Switch_0* sets the ML-based intrusion detector to training mode.
 - The output from the LLM-based unknown IoT traffic analysis module is fed into the ML-based intrusion detector to perform knowledge distillation [50].
 - *Switch_1* routes the output from the LLM-based unknown IoT traffic analysis module to be the node's final output during this phase.

Finally, the final output determined by the node in either phase is uploaded to the LLM-QBc-DB. Fig. 8 shows the working procedure of the dual-switch mechanism-based adaptive model updater. The pseudocode for this mechanism is detailed in Algorithm 2 in Appendix A.

4.4.4. LLM-QBc-DB node management module

LLM-based QBc authority agent (LLM-QBc-AA). The LLM-based post-quantum secure blockchain authority agent comprises two primary components: a credit score calculation algorithm and an LLM agent. The credit score calculation algorithm can calculate a credit score for each ordinary node based on its behavior consistency with other nodes and historical behavior. Then the LLM agent leverages these credit scores to determine whether an ordinary node has been corrupted. In summary the LLM-based post-quantum secure blockchain authority agent work in following procedure.

1. The agent first executes the Credit Score Calculation and Collusion Detection Algorithm. This algorithm continuously computes a dynamic credit score for each ordinary node based on its behavioral consistency and generates specific alerts for suspicious activities like collusion.
2. When a node is flagged by the algorithm (e.g., due to a zero credit score or a collusion alert), the system automatically compiles a comprehensive judgment case. This case file aggregates the score, its recent trend, and other contextual data, preparing it for the final decision.
3. Finally, the LLM agent receives and analyzes the contextual case file to make the ultimate governance decision. It leverages its reasoning capabilities to differentiate between coordinated malicious patterns and transient network anomalies, issuing a definitive command to either block the node or continue monitoring it.

Beyond straightforward credit scoring, the credit score calculation algorithm incorporates conditional logic to address specific malicious activities, such as identifying patterns indicative of collusion attacks among a group of nodes. Once these attacks are identified, a corresponding alert is also sent to the LLM agent. As a result, the cornerstone of the authority agent's governance capability is the Credit Score Calculation Algorithm. This algorithm is designed to quantify the trustworthiness of each ordinary node by assessing its behavior from two perspectives: its consistency with the network mainstream and its deviation from its own historical patterns. The following section details its procedural steps and mathematical formulation.

The credit score calculation algorithm. The credit score calculation algorithm dynamically assigns and updates a credit score for each ordinary node. Initially, each node has a predefined credit score $CS_i = 100$. In every broadcast cycle, a node's submitted input is processed into a normalized feature vector, denoted as V_i . This process involves averaging if multiple inputs are provided and employing embedding textual data by FastText techniques. A 'mainstream vector', V_M , is then determined for each type of node by calculating the credit-score-weighted average of the input vectors V_j from all nodes j belonging to that specific type, using the formula:

$$V_M = \frac{\sum_{j \in \text{Type}(i)} (CS_j \cdot V_j)}{\sum_{j \in \text{Type}(i)} CS_j} \quad (2)$$

Subsequently, two key metrics are computed for each node i . The first is the 'consistency with other nodes score', $S_{\text{consistency_others_i}}$. This score is derived by calculating the L2 distance $D_{i,M} = \|V_i - V_M\|_2$ between the node's input vector V_i and the mainstream vector V_M . The reciprocal of this distance, $1/D_{i,M}$, is then min-max normalized across all nodes of the same type to a range of $[-1, 1]$ to yield $S_{\text{consistency_others_i}}$.

$$\text{value}_{i,M} = \frac{1}{D_{i,M}} \quad (3)$$

$$\text{value_set}_M = \{\text{value}_{k,M} \mid i \in \text{Nodes} \wedge \text{Type}(k) = \text{Type}(i)\} \quad (4)$$

$$S_{\text{consistency_others_i}} = \frac{\text{value}_{i,M} - \min(\text{value_set}_M)}{\max(\text{value_set}_M) - \min(\text{value_set}_M)} \times 2 - 1 \quad (5)$$

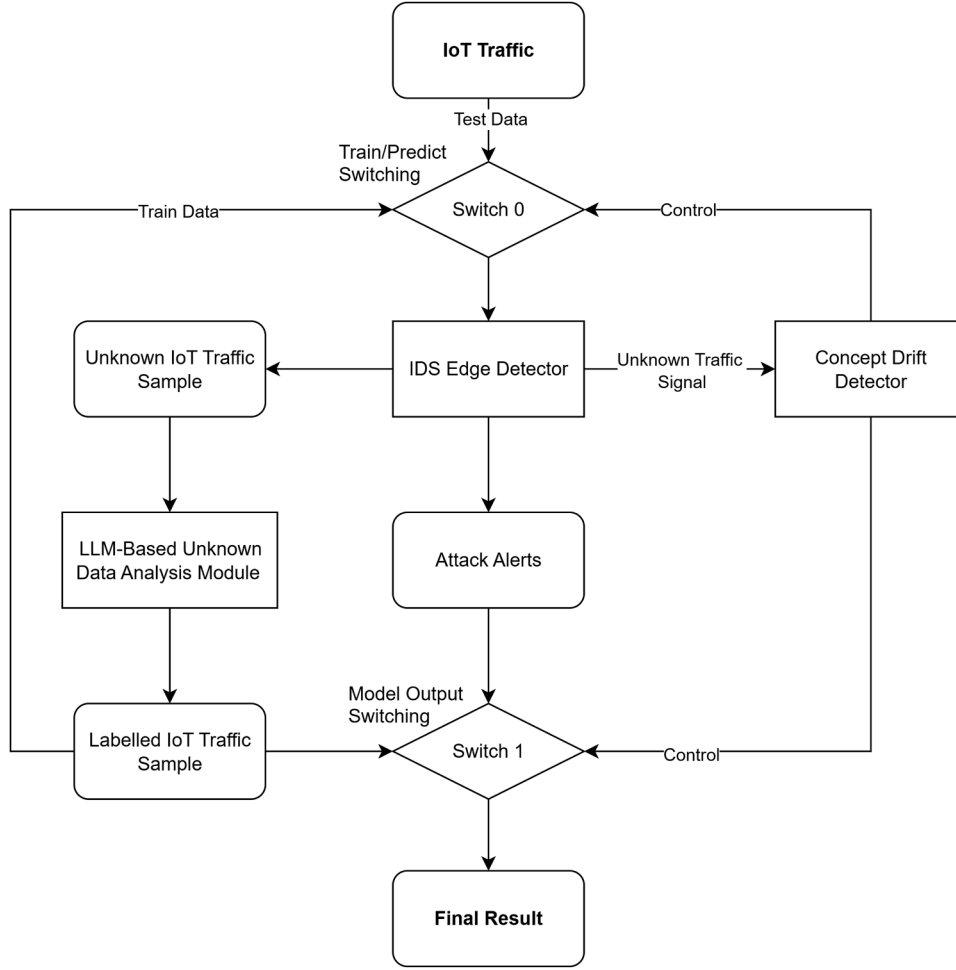


Fig. 8. Working procedure of dual-switch mechanism-based adaptive model updater.

The second metric is the ‘consistency with historical input coefficient’, C_{history_i} . This is obtained by computing the L2 distance $D_{i,H} = \|V_i - V_{H_i}\|_2$ between V_i and the node’s own average historical input vector V_{H_i} . This distance $D_{i,H}$ is then min-max normalized across all ordinary nodes to a range of $[0, 1]$ to yield C_{history_i} .

$$\text{value}_{i,H} = D_{i,H} \quad (6)$$

$$\text{value_set}_H = \{\text{value}_{k,H} \mid i \in \text{Nodes} \wedge \text{Type}(k) = \text{Type}(i)\} \quad (7)$$

$$C_{\text{history}_i} = \frac{\text{value}_{i,H} - \min(\text{value_set}_H)}{\max(\text{value_set}_H) - \min(\text{value_set}_H)} \quad (8)$$

The change in the node’s credit score for the current cycle, ΔCS_i , is the product of these two metrics:

$$\Delta CS_i = S_{\text{consistency_others}_i} \times C_{\text{history}_i} \quad (9)$$

Finally, the node’s credit score CS_i is updated by adding this ΔCS_i , while the range of CS_i is limited to $[0, 200]$.

$$CS_{i,t+1} = \min(\max(CS_{i,t} + \Delta CS_i, 0), 200) \quad (10)$$

where t is the index blockchain broadcast time period. Additionally, the algorithm assesses collective input consistency among low-credit-score nodes ($CS_i < 60$, the threshold is determined by grid search via

multiple experiments) after each update cycle, dispatching a collusion attack alert to the LLM agent if an unusually high similarity indicative of coordinated malicious behavior is detected. The detailed procedural steps of this algorithm are outlined in [Algorithm 1](#).

Meanwhile, the LLM agent operates based on a prompt word template to interpret the results from the credit score calculation algorithm. It is responsible for executing the final decision to block an ordinary node if its credit score drops to zero or if it is flagged as participating in a collusion attack according to the result of [Algorithm 1](#). [Fig. 9](#) shows how this QBc authority agent works.

Algorithm 1 Credit Score Calculation and Collusion Detection Algorithm.

Input:

For each ordinary node i in Nodes:

Input _{t} : Current input data from node i for the current broadcast cycle t .

CS_i : Current credit score of node i .

H_i : List of historical input vectors for node i .

Output:

For each ordinary node i in Nodes:

CS'_i : Updated credit score of node i .

- 1: Initialize collections: All_D_H $\leftarrow$ empty list; for each node type T , All_Recip_D_M_T $\leftarrow$ empty list.
- 2: Collusion_Alert $\leftarrow$ False.
- 3: Current_Cycle_Vectors $\leftarrow$ empty map.

```

4: for all node  $i$  in Nodes do
5:    $V_i \leftarrow \text{ProcessInput}(\text{Input}_i) \triangleright$  Convert to normalized vector (incl.
     FastText, averaging)
6:    $\text{Current\_Cycle\_Vectors}[i] \leftarrow V_i$ 
7:    $V_{H_i} \leftarrow \text{AverageVectors}(H_i) \triangleright$  Handle empty  $H_i$  with a default
8:    $D_{i,H} \leftarrow \text{L2Distance}(V_i, V_{H_i})$ 
9:   Add  $D_{i,H}$  to All_D_H.
10:  Store  $(V_i, CS_i, \text{Type}(i))$  for mainstream calculation.
11: end for
12: for all node type  $T$  do
13:    $V_{M_T} \leftarrow \frac{\sum_{j \in T} (CS_j \cdot \text{Current\_Cycle\_Vectors}[j])}{\sum_{j \in T} CS_j} \triangleright$  Using data from nodes of
     type  $T$ 
14:   for all node  $i$  in Nodes of type  $T$  do
15:      $D_{i,M} \leftarrow \text{L2Distance}(\text{Current\_Cycle\_Vectors}[i], V_{M_T})$ 
16:     Add  $(1/D_{i,M})$  to All_Recip_D_ $M_T$ .  $\triangleright$  Handle  $D_{i,M} \approx 0$  case
17:   end for
18: end for
19: Updated_CS  $\leftarrow$  empty map.
20: for all node  $i$  in Nodes do
21:   Let  $T_i$  be the type of node  $i$ .
22:    $S_{\text{consistency\_others}_i} \leftarrow \text{MinMaxNormalize}(1/D_{i,M}, \text{All\_Recip\_D\_}M_{T_i}, -1, 1)$ 
23:    $C_{\text{history}_i} \leftarrow \text{MinMaxNormalize}(D_{i,H}, \text{All\_D\_H}, 0, 1)$ 
24:    $\Delta CS_i \leftarrow S_{\text{consistency\_others}_i} \times C_{\text{history}_i}$ 
25:    $CS_{\text{temp}_i} \leftarrow CS_i + \Delta CS_i$ 
26:    $CS'_i \leftarrow \min(\max(CS_{\text{temp}_i}, 0), 200)$ 
27:   Updated_CS[ $i$ ]  $\leftarrow CS'_i$ 
28:   Append Current_Cycle_Vectors[ $i$ ] to  $H_i$ .
29: end for
// Collusion Attack Detection
30: Low_Credit_Score_Nodes  $\leftarrow []$ 
31: for all node  $i$  in Nodes do
32:   if Updated_CS[ $i$ ] < Low_CS_Threshold then
33:     Add  $i$  to Low_Credit_Score_Nodes
34:   end if
35: end for
36: if count(Low_Credit_Score_Nodes)  $\geq$ 
     Min_Low_CS_Nodes_For_Collusion_Check then
37:   Low_CS_Node_Input_Vectors  $\leftarrow []$ 
38:   for all node_index in Low_Credit_Score_Nodes do
39:     Add Current_Cycle_Vectors[node_index] to
     Low_CS_Node_Input_Vectors
40:   end for
41:   if count(Low_CS_Node_Input_Vectors) > 1 then
42:     Centroid  $\leftarrow \text{AverageVectors}(\text{Low\_CS\_Node\_Input\_Vectors})$ 
43:     Sum_Dist  $\leftarrow 0$ 
44:     for all vec in Low_CS_Node_Input_Vectors do
45:       Sum_Dist  $\leftarrow \text{Sum\_Dist} + \text{L2Distance}(\text{vec}, \text{Centroid})$ 
46:     end for
47:     Avg_Dist  $\leftarrow \text{Sum\_Dist} / \text{count}(\text{Low\_CS\_Node\_Input\_Vectors})$ 
48:     if Avg_Dist < Collusion_Consistency_Threshold then
49:       SendAlertToLLM("Potential collusion attack detected.")
50:       Collusion_Alert  $\leftarrow \text{True}$ 
51:     end if
52:   end if
53: end if
54: return Updated_CS, Collusion_Alert.

```

Design of robustness mechanisms against data poisoning, collusion, and quantum attacks. The defense mechanisms against various attacks in LQB-IDS are summarized as follows:

- **Data Poisoning Attacks:** Data poisoning attacks are prevented by the properties of blockchain. Meanwhile, the blockchain authority

Algorithm 2 Dual-Switch Mechanism-based Adaptive Model Updater.

Input:

iot_traffic: A continuous stream of incoming IoT traffic packets.

Output:

final_result: The detection result for each traffic packet.

```

// Initialization
1: ML_Detector  $\leftarrow \text{InitializeLightweightMLModel}()$ 
2: LLM_Analyzer  $\leftarrow \text{InitializeLLMAnalyzerModule}()$ 
3: ConceptDriftDetector  $\leftarrow \text{InitializeKSWINDetector}()$ 
4: system_state  $\leftarrow \text{NORMAL\_OPERATIONAL\_PHASE}$ 
// Main System Loop
5: while true do
6:   current_traffic  $\leftarrow \text{GetNextTrafficPacket}()$ 
7:   unknown_traffic_signal  $\leftarrow$ 
     IDS_Edge_Detector.getUnknownTrafficSignal(current_traffic)
8:   drift_detected  $\leftarrow \text{ConceptDriftDetector.check}(\text{unknown\_traffic\_signal})$ 
9:   if drift_detected and system_state ==
     NORMAL_OPERATIONAL_PHASE then
10:     system_state  $\leftarrow \text{MODEL\_UPDATE\_PHASE}$ 
11:   else if ML_Detector.isUpdateComplete() and system_state ==
     MODEL_UPDATE_PHASE then
12:     system_state  $\leftarrow \text{NORMAL\_OPERATIONAL\_PHASE}$ 
13:   end if
14:   if system_state == NORMAL_OPERATIONAL_PHASE then
// Normal Operational Phase
15:     switch_0  $\leftarrow \text{INFERENCE\_MODE}$ 
16:     switch_1  $\leftarrow \text{OUTPUT\_FROM\_ML\_DETECTOR}$ 
17:     final_result  $\leftarrow \text{ML\_Detector.predict}(\text{current\_traffic})$ 
18:   else if system_state == MODEL_UPDATE_PHASE then
// Model Update Phase
19:     switch_0  $\leftarrow \text{TRAINING\_MODE}$ 
20:     switch_1  $\leftarrow \text{OUTPUT\_FROM\_LLM\_ANALYZER}$ 
21:     llm_analysis_result  $\leftarrow \text{LLM\_Analyzer.analyze}(\text{current\_traffic})$ 
22:     final_result  $\leftarrow \text{llm\_analysis\_result} \triangleright$  Use LLM output for
     real-time decision
23:     ML_Detector.retrainViaKnowledgeDistillation
     (llm_analysis_result)
24:   end if
25:   BroadcastToBlockchain(final_result)
26: end while

```

agent can automatically block malicious attack information source nodes or compromised IoT device nodes according to credit score. This ensures that data in the blockchain is reliable.

- **Collusion Attacks:** These attacks are also detected by the credit score calculation algorithm. A collusion attack alert will be triggered if a group of ordinary nodes with low credit scores has an unusually high consistency in some period, and the related nodes will be blocked.
- **Quantum Attacks:** LQB-IDS is designed to be resilient against quantum attacks by using post-quantum secure cryptographic algorithms, such as quantum key generation in its blockchain.

4.5. Computational complexity and efficiency

During the normal operational phase, LQB-IDS uses an ML-based intrusion detector to do the end-to-end prediction. ML-based intrusion detectors use an RF model, which has a computational complexity of $O(N_t \times D_t)$, where N_t is the number of trees and D_t is the average tree depth. This low complexity ensures the feasibility of deployment on resource-constrained IoT edge devices.

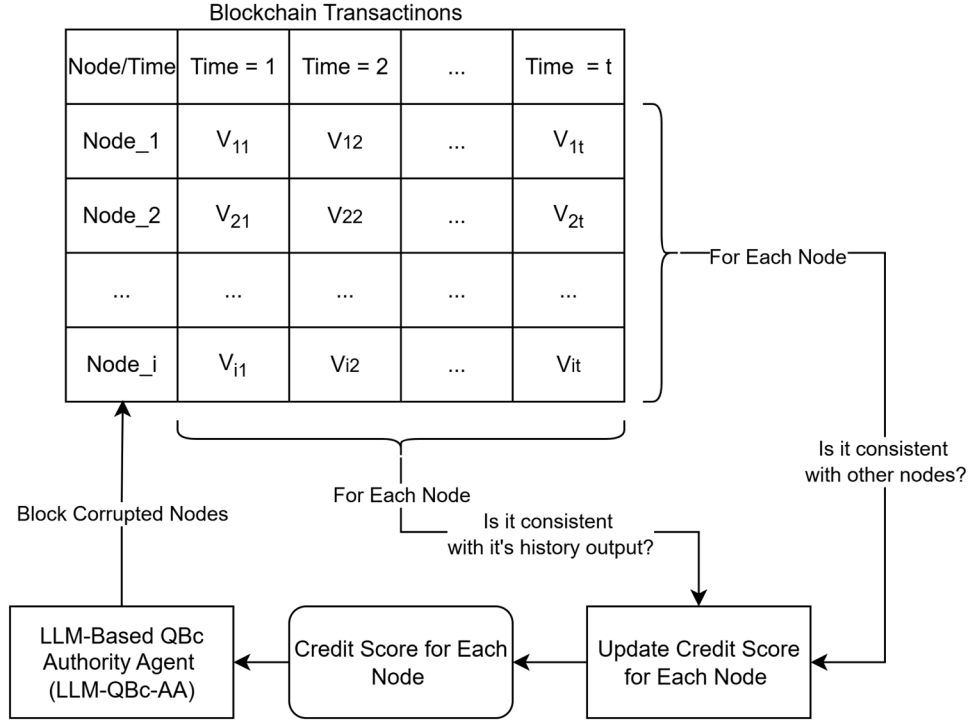


Fig. 9. Working procedure of LLM-based QBc authority agent.

Meanwhile, for the credit score calculation and collusion detection algorithm (Algorithm 1), the computational complexity is $O(n \times t \times d)$. Where n represents the number of ordinary nodes, t is the current broadcast cycle number (reflecting the growing size of the historical data), and d is the dimension of the feature vector. This means the algorithm's processing time per cycle grows linearly with the number of nodes and the length of the system's operational history.

However, a detailed computational complexity analysis of the model update phase is beyond the scope of this paper because its detection delay is predominantly governed by the inference time of commercial closed-source LLMs, for which a complexity analysis is not feasible due to being closed-source.

To bridge the preceding system model with the subsequent empirical evaluation, we now summarize the operational flow of LQB-IDS. The process begins with the IDS Edge Detector performing preliminary analysis on IoT traffic. Suspected unknown samples are then securely uploaded to the cloud via the LLM-QBc-DB. In the cloud, the LLM-Based Unknown IoT Traffic Analysis Module autonomously analyzes these samples. When concept drift is detected, the Dual-Switch Mechanism-based Adaptive Model Updater is triggered to retrain the edge model using knowledge distillation, ensuring rapid adaptation to new threats. Validating such a multi-layered, dynamic architecture necessitates a structured experimental design, a principle emphasized when discussing complex systems like quantum-secure networks [51]. Accordingly, the following section details the numerical experiments designed to assess the performance of LQB-IDS.

5. Numerical experiments and results

5.1. Experiment design

5.1.1. Hybrid IoT intrusion dataset with concept drift

In this paper, a specialized hybrid intrusion dataset with concept drift was designed to evaluate the performance of various IoT IDSs in adaptively detecting unknown attacks and adapting to concept drift within dynamic IoT network environments. This hybrid dataset was constructed by combining two distinct open-source IoT intrusion datasets:

the Canadian Institute for Cybersecurity Internet of Things Dataset 2023 (CIC-IoT-2023) [45] and the Canadian Institute for Cybersecurity Internet of Medical Things Dataset 2024 (CIC-IoMT-2024) [52].

The hybrid dataset is separated into three stages, with each stage including one-third of the overall dataset. Both normal and attack data from original datasets were sampled to maintain a consistent proportion in each stage.

- **Stage 1:** Only includes benign and attack data from the CIC-IoT-2023 dataset.
- **Stage 2:** Besides benign and attack data from the CIC-IoT-2023 dataset, benign data from CIC-IoMT-2024 is also introduced in this stage. These unseen benigns from CIC-IoMT-2024 are used to simulate concept drift in the IoT network environment. For example, new protocols are applied, like the Internet Group Management Protocol (IGMP), which is present in CIC-IoMT-2024 but absent in CIC-IoT-2023.
- **Stage 3:** Adds the attack data from the CIC-IoMT-2024 dataset to simulate unknown attacks, considering that the initial IDS models are to be trained only on data from CIC-IoT-2023.

By comparing the detection performance of different models from Stage 1 to Stage 2, their adaptability to environmental concept drift can be observed. Afterwards, the change in detection performance from Stage 2 to Stage 3 allows for the assessment of the adaptive detection capabilities against unknown attacks of each model. Fig. 10 illustrates the data source proportions across each stage.

5.1.2. Collusion attack scenario setup

In this paper, three typical attack scenarios are primarily considered: data poisoning attacks, collusion attacks, and quantum attacks. The defense against data poisoning and quantum attacks predominantly relies on the inherent properties of the post-quantum secure blockchain (QBc) itself. The specific defense mechanisms and implementation details of such a blockchain are based on the work by Xu et al. [43], which LQB-IDS adapted for its LLM-QBc-DB.

Collusion attacks, however, are specifically addressed by the proposed LLM-based post-quantum secure blockchain authority agent and

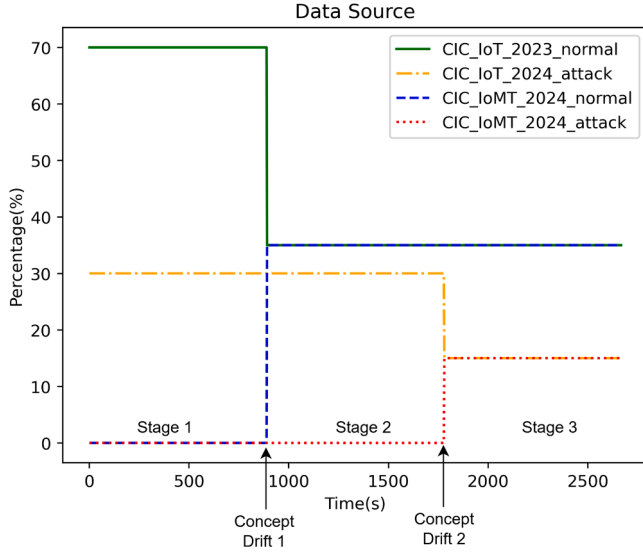


Fig. 10. Proportion of data sources versus time.

its embedded credit score calculation and collusion detection algorithm. To validate the efficacy of this agent and its algorithm in identifying and mitigating collusion attacks, the following experimental setup was designed.

In this experiment, the behavior of 10 IoT edge nodes (ordinary nodes), designated as Node 0 through Node 9, was simulated over 1000 blockchain broadcast cycles. These nodes were categorized into three types to represent diverse network conditions and behaviors:

- 1. Normal Nodes:** These nodes are exposed to the main IoT network and can perceive legitimate network traffic as well as potential attack data. They are not compromised by attackers and are assumed to faithfully upload the detection results from their local IDS edge detector and DL-based unknown IoT traffic detector to the blockchain.
- 2. Isolated Normal Nodes:** These nodes represent devices operating within a secure, isolated subnet. While also loyal and uncompromised, they are not reachable by attackers and thus do not encounter malicious traffic. Consequently, they consistently report results indicating no detected attacks.
- 3. Corrupted Nodes:** These nodes have been compromised and are under the control of an attacker. They can be manipulated to send arbitrary or malicious data to the blockchain, deviating from the true outputs of their local detection modules.

The same data set was fed into all these nodes except isolated normal nodes. For the data fed into the isolated normal nodes, the attack data was filtered. Two distinct sets of experiments were conducted to evaluate the system's robustness:

- **Scenario 1: Non-Collusive Malicious Behavior:** In this scenario, the corrupted nodes are programmed to send erroneous, random data to the blockchain. This setup aims to test the credit score algorithm's ability to identify and penalize individual nodes exhibiting abnormal behavior without coordinated effort.
- **Scenario 2: Collusion Attack:** In this scenario, the corrupted nodes actively collude. They coordinate to send consistent, specifically crafted malicious data to the blockchain. The objective of this coordinated attack is to manipulate the 'mainstream vector' calculated by the Credit Score Calculation and Collusion Detection Algorithm, thereby attempting to illegitimately maintain or boost their credit scores or disrupt the scoring of legitimate nodes.

The comparison of the LLM-based post-quantum secure blockchain authority agent's performance under these two scenarios allows for the

assessment of its capability to handle general misbehavior as well as to detect and counteract sophisticated collusion attempts. The outcomes of these experiments are demonstrated and discussed in Section 5.2.3.

5.1.3. Data preprocessing

Data preprocessing involves four main steps: data cleaning, feature selection, data normalization, and two-dimensional (2D) feature transformation (for DCAE only). Initially, data cleaning was conducted to remove entries with missing values, duplicate records, and any obvious outliers from the dataset. Following data cleaning, feature selection was performed using the SelectKBest [53] method from the scikit-learn library. After the relevant features had been selected, data normalization was applied to these selected features using the StandardScaler method. Finally, for the DCAE model, the two-dimensional (2D) feature transformation was performed using the Triangle Area Maps (TAMs) method [54]. The detailed steps of the TAM method are as follows: For a d -dimensional data vector $\mathbf{x} = (x_1, x_2, \dots, x_d)$, a $d \times d$ matrix $\mathbf{Y}$ is constructed. Each element y_{ij} in this matrix is derived from the input vector $\mathbf{x}$ using the equation:

$$y_{ij} = x_i \times x_j, \quad \text{for } i \in [1, d], j \in [1, d] \quad (11)$$

This transformation results in a $d \times d$ matrix for each 1D input vector, which serves as the input for the DCAE.

5.1.4. Post-quantum-secure blockchain and LLM implementation

The Post-Quantum-Secure Blockchain (QBC) is implemented directly based on an open-source project [43,55,56], which applies a digital signature scheme based on the Dilithium algorithm. The core post-quantum secure cryptographic functionalities within this framework were implemented primarily through the pqcrypto library [57].

The LLMs in this study were accessed via their respective commercial application programming interfaces (APIs). The performance of several prominent models, including Generative Pre-trained Transformer 4 Omni (GPT-4o) [58], Meta Large Language Model 4 Maverick/Scout (Llama 4 Maverick/Scout) [59], Gemini 2.5 Pro [60], and DeepSeek Version 3 (DeepSeek-V3) [61], was evaluated. Based on this comparative evaluation, GPT-4o demonstrated superior performance in our experimental setup. Thus, it was consequently selected as the core LLM for the LQB-IDS framework.

5.1.5. Configuration and implementation details

To ensure the reproducibility of our experimental setup, this section details the key configuration parameters for both the DCAE model and the blockchain framework.

For the DCAE model training, we set a maximum of 50 epochs with a batch size of 32. The Adam optimization algorithm was utilized for weight updates. To prevent overfitting, a ReduceLROnPlateau learning rate scheduler was employed, which reduces the learning rate by a factor of 0.5 if validation loss does not improve for two consecutive epochs. Furthermore, an early stopping mechanism with a patience of 5 epochs was implemented.

The blockchain component was configured to operate in an IoT-friendly environment. We used the Dilithium2 algorithm for post-quantum secure signatures in a network of 10 nodes. New blocks were created in an event-driven manner, triggered by successful model aggregation cycles. A comprehensive summary of all hyperparameters and configuration settings for both components is provided in Table 3.

5.1.6. Computational resource usage

As the detection delay is influenced by the computational resource used for the simulation experiment, and to increase the reproducibility of this study, Table 4 specifies the computational resources employed for all conducted experiments.

Table 3
Hyperparameter and blockchain configuration settings.

DCAE Model Training		
Parameter	Setting	Value
Optimizer	Adam	NA
Learning Rate	Learning Rate with Scheduler	$lr = 10^{-3}$
Learning Rate Scheduler	ReduceLROnPlateau	mode=min, patience=2, factor=0.5
Early Stopping	Yes	max_epochs=50, patience=5
Batch Size	NA	32
Dual-Switch Trigger	KSWIN Algorithm	α (Significance Level)=0.005
Post-Quantum-Secure Blockchain Configuration		
Parameter	Setting	Value
PQC Signature Scheme	Dilithium2	NA
Total Nodes in Experiment	NA	10
Block Interval	Event-Driven	NA
Block Size	Depand on a event-driven dynamic policy	Less than 10,000 package per block

Table 4
Computational resource for the experiment.

Hardware	Information
CPU	Intel Core i7-13700H @ 2.4 GHz
Memory	32 GB @ 5200 MHz
GPU	NVIDIA RTX 4060 8 GB
Disk	Phison PCIe4.0 E21 512 GB SSD

5.1.7. Evaluation metrics

Four key evaluation metrics are used to compare the performance of different IoT IDSs:

- Detection Rate (DR):** DR measures the proportion of actual attacks that the IDS correctly identifies.
- False Positive Rate (FPR):** FPR indicates the proportion of benign instances that are incorrectly classified as malicious attacks. This metric is particularly important for the practical applicability of IDSs, as a high FPR can overwhelm analysts with false alarms, reducing overall system efficiency and trustworthiness.
- F1-Score:** The F1-Score is the harmonic mean of precision and recall (sensitivity), providing a single score that balances both metrics, especially useful for imbalanced datasets where DR alone might be misleading.
- Detection Delay:** This refers to the time taken by the IDS to identify and report a security threat from the moment the IoT traffic reaches the IDS.

The experimental results presented are the average of 10 independent repetitions. A paired *t*-test was conducted on these 10 sets of results, yielding a *p*-value of 0.048, indicating a statistically significant difference in the observed performances. To emphasize the importance of not relying on labeled data, only 50 % of the labeled data was provided during the model updating stage, demonstrating the model's robustness and effectiveness in self- or semi-supervised scenarios.

5.2. Results and analysis

5.2.1. Detection rate, false positive rate and F1-score

To evaluate the performance of our proposed LQB-IDS, a comparative analysis was conducted against several recent IoT IDSs: IDS-Agent [32], AILL-IDS [24], CAEAID [18], HFIN [20], HuntGPT [31], SecurityBERT [30], BiGRU-LSTM-Attention [27] and SD-PQBA [36]. These IoT IDSs are introduced in the literature review section and were chosen as benchmarks because they each signify distinct and advanced approaches within the domain of dynamic IoT IDS development. IDS-Agent is representative of the increasing trend towards leveraging large language models for enhanced analytical capabilities and explainable

AI in detecting IoT threats. AILL-IDS is designed for automatic and continuous incremental learning, which is critical for adaptation in highly dynamic network environments. Furthermore, CAEAID is selected for its sophisticated framework that employs incremental contrastive learning, addressing the pivotal challenges of concept drift and evolving data complexities in IoT. Lastly, HFIN represents a comprehensive approach to managing ever-changing threats within Industrial IoT contexts through a hierarchical architecture that effectively integrates federated and class-incremental learning techniques. Table 5 compares the overall performance of LQB-IDS with benchmark IoT IDSs.

Table 5 shows that LQB-IDS exhibited the best performance across all evaluated metrics, followed by IDS-Agent and BiGRU-LSTM-Attention, while CAEAID and HFIN showed comparatively modest results. These observations can be primarily explained by the differences in each IoT IDS design. As discussed in Section 5.1, the labeled data available for model updating was restricted in our experiment. Consequently, LQB-IDS and IDS-Agent, which leverage LLMs for dynamic data analysis, are less reliant on labeled data for their core model updating processes, thus achieving relatively strong performance. Similarly, other LLM-based models like HuntGPT and SecurityBERT achieve competitive detection results, but as the table indicates, they do so at the cost of significantly higher inference times and detection delays, which highlights a common trade-off in end-to-end LLM architectures that our hybrid approach effectively mitigates. The reason for the better performance of LQB-IDS compared to IDS-Agent is that LQB-IDS applies the well-designed DCAE model, which possesses more advanced analyzing capabilities for complex IoT traffic. In contrast, IDS-Agent employs a voting mechanism based on traditional ML models for processing IoT traffic. Similarly, AILL-IDS achieved commendable performance as it applies a semi-supervised learning strategy, enabling efficient model updates even with a limited amount of labeled data. It is also worth noting that while SD-PQBA pioneers the use of post-quantum cryptography, its detection performance metrics do not reach the top tier, underscoring the challenge of balancing advanced security with high-performance intrusion detection—a gap that LQB-IDS successfully bridges. Finally, as CAEAID and HFIN lack designs to avoid relying on the labeled data, these IDSs had comparatively lower overall performance.

To better observe the performance of different IoT IDSs when concept drift and new attacks occur, Figs. (11–13) illustrate the trends of Detection Rate (DR), False Positive Rate (FPR), and F1-Score over time for the evaluated IoT IDSs, respectively.

From Figs. (11–13), a consistent trend is observed across all evaluated IoT IDSs. Generally, the performance of all IDSs, including DR, FPR, and F1-Score, was highest in Stage 1, followed by Stage 2, and then Stage 3. The reason for these decreases is the increasing complexity and diversity of the data across the stages. For example, stage 1 exclusively utilizes data from the CIC-IoT-2023 dataset, while stage 2 introduces

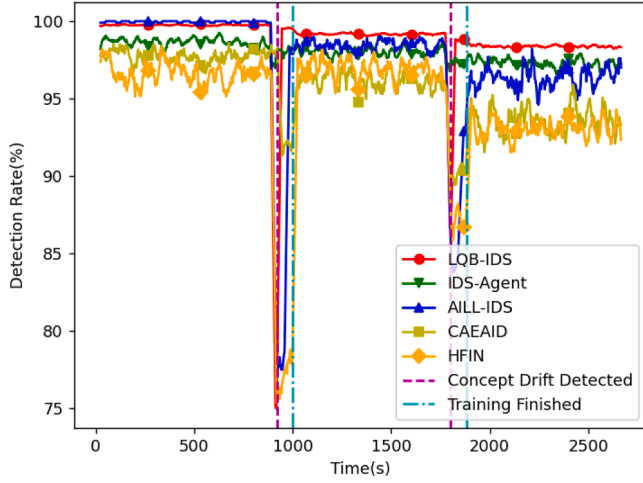


Fig. 11. Detection rate (%) of different IoT IDSs over time.

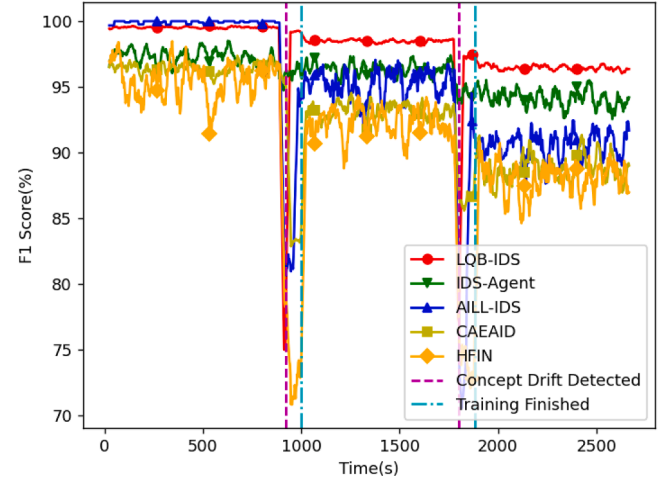


Fig. 13. F1 score (%) of different IoT IDSs over time.

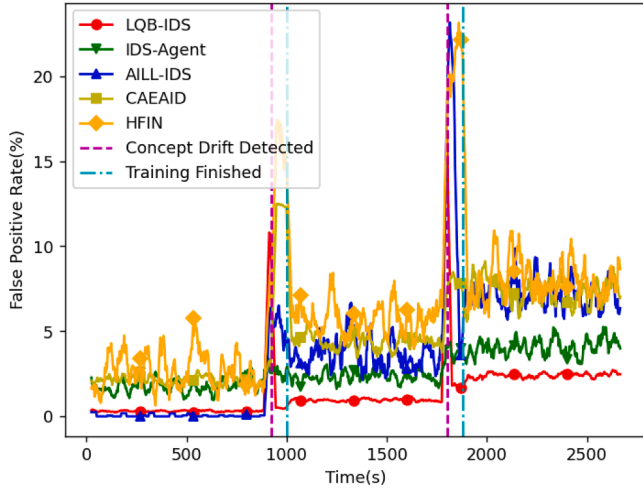


Fig. 12. False positive rate (%) of different IoT IDSs over time.

normal traffic from the CIC-IoMT-2024 dataset, presenting new patterns of benign behavior that can challenge the established models. Finally, Stage 3 introduces both normal and attack data from the CIC-IoMT-2024 dataset, introducing novel attack vectors and further diversifying the data distribution. This progressive increase in data complexity and the number of classes naturally leads to a decline in detection performance for most systems as they adapt.

Another critical observation pertains to the performance dynamics during the transition periods, specifically between the detection of concept drift (“Concept Drift Detected” marker) and the completion of model retraining (“Training Finished” marker). During these intervals:

IDS-Agent exhibited a relatively stable performance without significant decay. This is because IDS-Agent continuously leverages LLMs for dynamic data analysis, allowing it to adapt to data changes more fluidly without a distinct performance drop induced by model updating. However, this reliance on LLM for real-time analysis typically incurs a higher detection delay, a factor that is discussed in detail in Section 5.2.2.

Among the IDSs with DL or ML models, LQB-IDS demonstrated the most rapid recovery from performance degradation. When concept drift is detected, LQB-IDS can immediately switch to LLM-based dynamic analysis, using the dual-switch learning mechanism. This interim reliance on the LLM sustains detection capabilities while the primary ML model is updated via knowledge distillation. Once the ML model is retrained, the system switches back. The brief performance dip observed is

primarily due to the slight detection delay of the concept drift detector itself.

Besides LQB-IDS, AILL-IDS showed the second-fastest recovery. It utilizes a dual ensemble learning strategy, where an auxiliary model undergoes incremental learning concurrently with the main model’s inference. When concept drift is detected, AILL-IDS can switch to the updated auxiliary model, facilitating a quicker adaptation than models requiring incremental learning from scratch.

In contrast, CAEAID and HFIN lack specialized mechanisms to increase the recovery speed during the model update phase. Consequently, their detection performance remained compromised for a longer period until their models are completely updated with the new data, leading to more performance loss during these critical transition windows.

5.2.2. Detection delay

Detection delay, which refers to the end-to-end time an IDS requires to analyze network traffic and generate an alert or result, which is a crucial metric for assessing its operational efficacy, is given in Table 5. It can be observed that LQB-IDS achieved the shortest average detection delay, whereas AILL-IDS, CAEAID, HFIN, BiGRU-LSTM-Attention, and SD-PQBA demonstrated moderate detection delays. In contrast, the detection delay of IDS-Agent, along with other LLM-based models like HuntGPT and SecurityBERT, was significantly higher than all other evaluated IoT IDSs. The superior performance of LQB-IDS in terms of low latency is attributed to its operational design. Model updates are conducted only in short periods, which only occupy a small fraction of the total runtime. During these periods, LQB-IDS employs LLM-based dynamic analysis and knowledge distillation to update the ML model. For the majority of the operational time, LQB-IDS relies on a lightweight ML model, enabling direct end-to-end inference and ensuring low detection delay. An additional experiment is also conducted to further prove the practical deployment feasibility on resource-constrained edge devices. In this experiment, the RF-based IDS Edge Detector was deployed on a Raspberry Pi 4B with 8 GB of memory. The memory consumption is recorded, which shows that the memory consumption is less than 50MB. Meanwhile, the detection delay increased to 51.146 ms per packet, which still remains acceptable. This result demonstrates the trade-off between computational overhead and detection delay: the limited computation resource will increase the detection delay, but the increased delay is still acceptable on a typical IoT edge device like Raspberry Pi 4B. As a result, LQB-IDS is well-suited for practical, real-world IoT environments, as this approach ensures minimal delay for the majority of its operation, leading to the lowest average detection delay.

It is critical to distinguish between the system’s real-time detection latency and the latency associated with its adaptive security updates via

Table 5
Comprehensive performance comparison on the CIC-IoT-2023 & CIC-IoMT-2024 hybrid dataset.

IDS Name	Detection Rate	False Positive Rate	F1-Score	Inference Time (ms per packet)	Detection Delay (ms per packet)	Detect Unknown Attacks
IDS-Agent [32]	0.9797	0.0272	0.9590	4.19	4.19	True
AILL-IDS [24]	0.9740	0.0371	0.9454	0.48	0.48	False
CAEAID [18]	0.9560	0.0478	0.9248	0.74	0.74	False
HFIN [20]	0.9450	0.0600	0.9065	0.42	0.42	False
HuntGPT [31]	0.9754	0.0590	0.9234	4.23	4.23	False
SecurityBERT [30]	0.9638	0.0270	0.9512	3.22	3.22	False
BiGRU-LSTM-Attention [27]	0.9788	0.0460	0.9388	0.71	0.71	False
SD-PQBA [36]	0.9525	0.0440	0.9273	0.73	0.83	False
LQB-IDS (Ours)	0.9871	0.0142	0.9772	0.16	0.23	True

the blockchain. The blockchain module operates asynchronously and is only triggered by the detection of unknown attacks. Therefore, it introduces no latency to the routine inspection of normal traffic. To quantify the overhead during these specific events, we measured the peak instantaneous latency. The PQC signing operation can take up to 5.31 ms, and a batch upload of aggregated threat data to the blockchain can take up to 14.18 ms. The most significant latency occurs during the retrieval of security policies, which can reach a peak of 697.61 ms. While this retrieval delay is substantial, it is an infrequent operation, occurring only when the system needs to update its defenses against a newly confirmed, unknown threat. This deliberate design choice creates a trade-off between accepting a temporary, high latency for a rare but critical security update and enabling the system's advanced adaptivity and resilience. By batching operations and isolating them from the primary detection pipeline, LQB-IDS ensures that its robust, future-proof security does not create a bottleneck in standard operational performance.

For AILL-IDS, CAEAID, and HFIN, all of which employ Deep Learning (DL) models, their detection delays were comparatively moderate. Among them, CAEAID utilizes a more complex model architecture, resulting in a relatively higher detection delay. Conversely, AILL-IDS applies a lightweight model design, and HFIN employs a federated learning approach that is also based on lightweight models, contributing to their relatively lower detection delays. Finally, the reason for IDS-Agent's longest detection delay is its continuous reliance on an LLM for dynamic data analysis. As LLMs are inherently large and computationally intensive, their inference times are considerably longer, leading to the much longer detection delay for IDS-Agent compared to the other systems.

Besides average detection delay, the instantaneous detection delay at each time point is also crucial, particularly for understanding how an IDS performs during dynamic events such as concept drifts and model updates. Fig. 14 illustrates the detection delay of each IoT IDS over time, providing insights into their real-time responsiveness.

Fig. 14 shows that, consistent with previous observations, IDS-Agent had a stably high detection delay throughout the experiment due to its continuous reliance on LLM-based dynamic analysis. For LQB-IDS, the delay remained low during its normal operational phases, as it utilizes a lightweight end-to-end ML model in this period. However, during the model update periods (between the "Concept Drift Detected" and "Training Finished" markers), a noticeable spike in detection delay occurred, which is due to the LLM-based dynamic analysis used in LQB-IDS in this period.

In the case of AILL-IDS, the significant increase in detection delay appeared during its update phases. This is due to its dual ensemble learning strategy: two models are inferring in parallel in this period. Conversely, CAEAID and HFIN showed a lesser increase in their detection delays during model updating periods. This is because they undergo standard incremental training or retraining processes without a specialized updating strategy to lessen the performance degradation in this period. Thus, as discussed in the previous section, this approach comes at the cost of a more substantial degradation in their detection performance

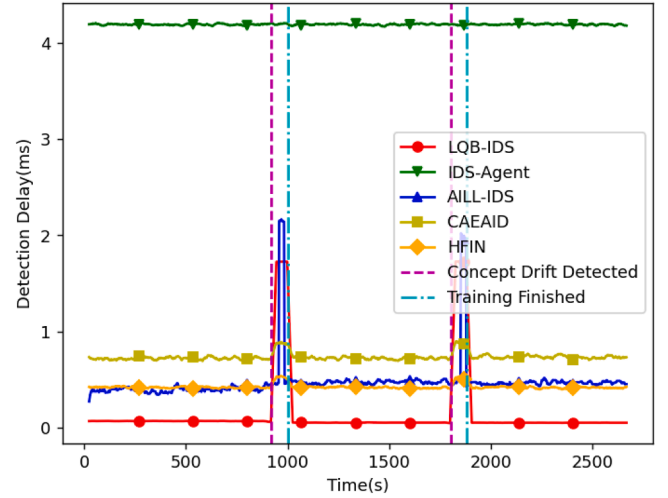


Fig. 14. Detection delay (ms) of different IoT IDSs over time.

(e.g., DR, F1-Score) during these periods. Finally, the somewhat more noticeable increase in HFIN's delay compared to CAEAID during updates might be linked to the computational overhead associated with its federated learning weight merging algorithm.

5.2.3. Ablation studies

To quantitatively isolate and validate the individual contributions of the core components within the LQB-IDS framework, a series of ablation studies were conducted. In this experiment the performance of the LQB-IDS model is completed against three degraded versions, each with one key component removed: the dual-switch mechanism, the LLM-based dynamic analysis module, and the Post-Quantum-Secure Blockchain (LLM-QBc-DB) module. The results are summarized in Table 6.

From Table 6, the following observations can be made:

- **Contribution of the Dual-Switch Mechanism:** Removing the dual-switch mechanism, which forces the system to rely solely on the LLM for dynamically analyzing all traffic, resulted in an increase in the average detection delay (from 0.23 ms to 1.84 ms), while the detection rate remained virtually unchanged. This result demonstrates the importance of the dual-switch mechanism for the system's efficiency.
- **Contribution of the LLM Module:** The removal of the LLM-based dynamic analysis module led to a decrease in the detection rate from 98.71 % to 69.83 %. This decline occurred because, without the LLM, the system relied solely on a static ML model for detection, which could not handle unknown attacks or concept drift introduced in our dataset. These results validate the LLM's role in dynamic analysis and adaptive detection.

Table 6
Ablation study of LQB-IDS components.

Model configuration	Detection rate	Average delay (ms per package)	Contribution validated
LQB-IDS (Full Model)	0.9871	0.23	Synergistic performance
w/o Dual-Switch (LLM + LLM-QBc-DB)	0.9867	1.84	Crucial for maintaining low latency and operational efficiency.
w/o LLM (only static ML model)	0.6983	0.21	Essential for detecting unknown attacks and ensuring high accuracy.
w/o LLM-QBc-DB (Dual-Switch + LLM without RAG)	0.9252	0.23	Provides trusted external knowledge that enhances the LLM's analytical accuracy.

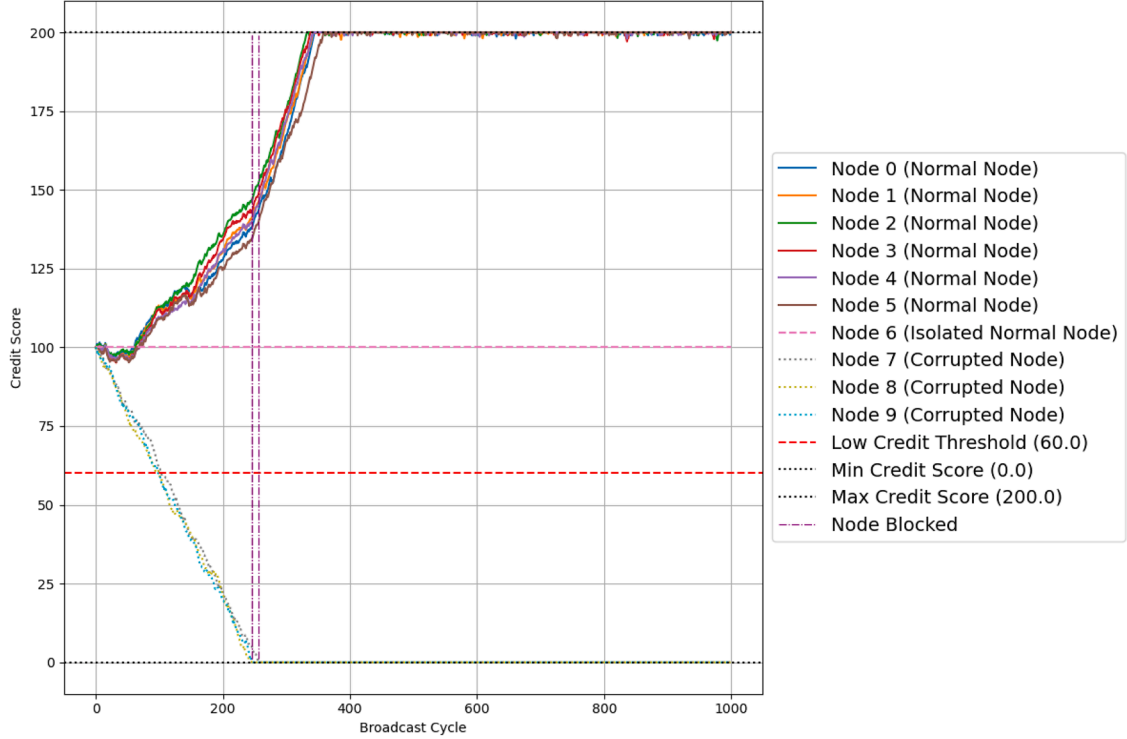


Fig. 15. Credit score evolution of nodes under non-collusive malicious behavior.

- **Contribution of the LLM-QBc-DB** When the LLM-QBc-DB module was removed, the system's detection rate dropped to 92.52%. This occurred because, without LLM-QBc-DB, the LLM analyzer could only rely on its pre-trained knowledge for analysis, which may not be sufficient to handle unknown attacks.

In summary, the ablation studies confirm that the components of LQB-IDS are not merely additive but highly synergistic. The dual-switch mechanism ensures efficiency, the LLM provides accuracy and adaptability, and the LLM-QBc-DB delivers data security and enhanced intelligence, collectively creating a robust and high-performance system.

5.2.4. Experimental validation of robustness against collusion attack

While previous discussions focus on the overall detection performance and efficiency of LQB-IDS compared to other systems, this section shifts the focus to the internal robustness of LQB-IDS itself, particularly its ability to handle scenarios involving corrupted nodes and coordinated collusion attacks. The primary goal is to demonstrate the system's capability to identify and subsequently block such malicious entities.

Figs. 15 and 16 illustrate the credit score evolution for each node under the scenarios described in Section 5.1.2. These figures also indicate the points at which nodes are blocked by the system due to their malicious activities.

For normal nodes (Nodes 0–5), their credit scores generally exhibited an upward trend with some fluctuations, eventually reaching and maintaining the maximum credit score of 200.0 in both experiments.

This trajectory is a result of how their behavior interacts with the credit score calculation algorithm (Algorithm 1). Firstly, these nodes constitute the majority and are exposed to the same dynamic IoT network environment, causing their submitted data to be largely consistent with each other and thus align closely with the 'mainstream vector.' This alignment yields a positive 'consistency with other nodes score.' Secondly, due to the evolving nature of the network traffic, the data reported by these normal nodes often differs from their own historical aggregated outputs. This divergence from past patterns leads to a higher 'consistency with historical input coefficient.' The change in credit score is equal to the 'consistency with historical input coefficient' times the 'consistency with other nodes score.' Since both these components are generally positive and significant for normal nodes, their product results in a positive value, causing the credit score to increase. Thus, normal nodes are never blocked by the system.

Regarding the isolated normal nodes (Node 6), it is observed in both experiments that their credit scores remained stable around their initial value. The reasons for this observation are detailed below: Although the outputs from these isolated nodes—typically reporting no attacks due to their isolated network environment—might be inconsistent with the 'mainstream vector' influenced by attack data from normal nodes, which may lead to a negative 'consistency with other nodes score.' However, as these nodes are in an isolated and stable environment, their current outputs are highly consistent with their own historical outputs, leading the 'consistency with historical input coefficient' to approach a value of zero. Consequently, when the change in credit score is calculated as

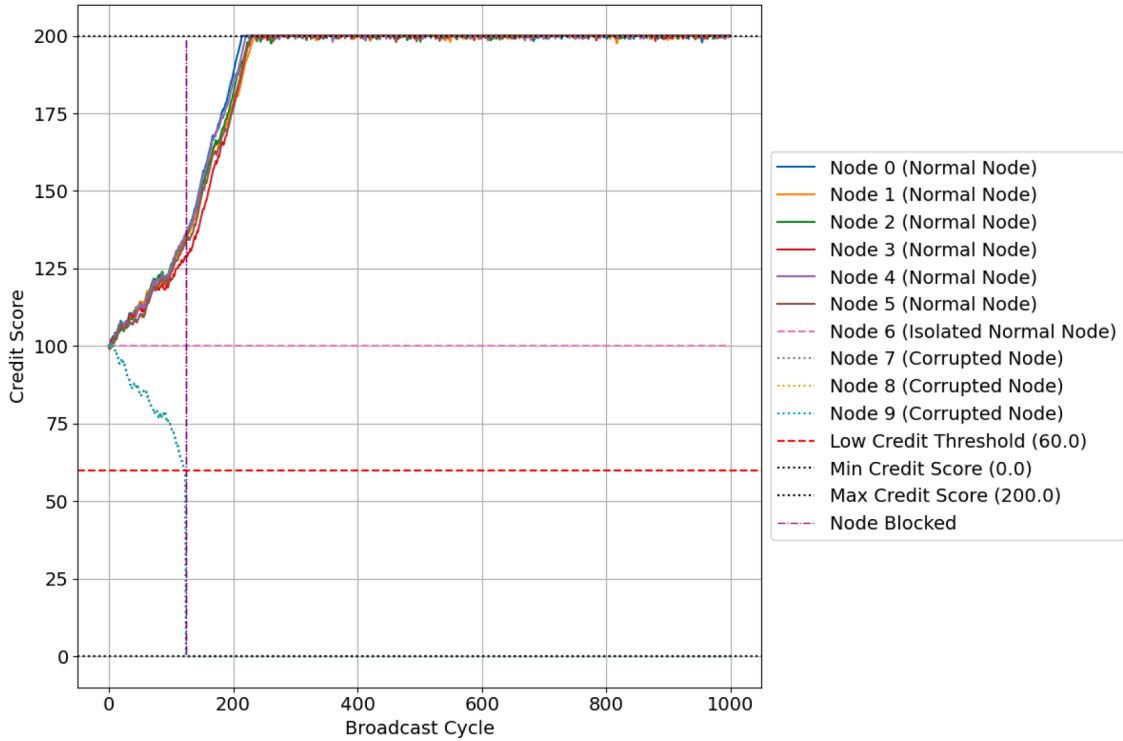


Fig. 16. Credit score evolution of nodes under collusion attack scenario.

the product of the ‘consistency with other nodes score’ and this near-zero ‘consistency with historical input coefficient,’ the resulting change is also close to zero. Thus, the credit scores remain at a constant level. Therefore, these consistently isolated and unattacked nodes, despite not actively contributing to the detection of ongoing network-wide threats, are not penalized by the system and are not blocked.

Finally, concerning the corrupted nodes (Nodes 7, 8, and 9 in both scenarios), their credit scores consistently decreased in both experiments, albeit through slightly different dynamics, ultimately leading to their blockage. This general downward trend occurs because their malicious or erroneous data submissions significantly deviate from the data patterns established by the majority of normal nodes, resulting in a negative ‘consistency with other nodes score.’ Additionally, their arbitrary or specifically crafted malicious inputs are typically inconsistent with legitimate historical data patterns, resulting in a significant ‘consistency with historical input coefficient.’ Thus, the final change in credit score is a non-zero negative value.

In the non-collusive malicious behavior scenario (Fig. 15), where corrupted nodes transmit individually random and erroneous data, their credit scores declined steadily. Since their malicious inputs are uncoordinated and individually distinct, the system does not detect high input consistency among these low-credit-score nodes. Therefore, no specific collusion alert was triggered. These nodes were eventually blocked when their credit scores diminish to the minimum threshold (0.0).

Conversely, in the collusion attack scenario (Fig. 16), where corrupted nodes coordinate to send consistent malicious data, the decline speed of their credit scores was slower compared to the non-collusive case. The reason for this observation is that their unified input slightly influences the mainstream vector, for example, making the mainstream vector more “close” to the attacker’s input, thereby making their ‘consistency with other nodes score’ less drastically negative at the very outset. However, as they remain a minority and their data is fundamentally malicious, their scores inevitably decreased. Then, once their credit scores fall into the low-credit range (below the 60.0 threshold as indicated in Algorithm 1 and visually in the figures), the high consistency among

their own malicious inputs is identified by the collusion detection mechanism embedded in Algorithm 1. This triggers a collusion attack alert, and these colluding nodes were blocked by the LLM agent. Thus, regardless of whether they act individually or in a coordinated collusion attack, corrupted nodes were effectively identified and blocked by the LQB-IDS within a relatively short period, proving the robustness of LQB-IDS against collusion attacks.

5.2.5. Interpretation of false positives and negatives and practical reliability

To comprehensively assess the practical reliability of the proposed system, we conducted a further analysis of the sources of false positives (FPs) and false negatives (FNs).

Our investigation reveals that the primary source of transient errors is not the LLM component. Our manual error analysis indicates that approximately 91.58% of these errors are attributable to the inherent detection latency of the KSWIN concept drift detector. This latency creates a brief window before control is switched to the LLM. Within this period, the un-updated edge ML model must classify traffic it cannot comprehend, which consequently leads to a temporary surge in FPs.

The remaining 8.42% of errors are related to the intrinsic weaknesses of the LLM, such as model hallucination or a failure in the RAG process. However, LQB-IDS was designed to proactively mitigate these risks. As detailed in Section 4.4.2, we enforce a strict output format via regular expressions to minimize hallucination. Furthermore, the RAG process is grounded in numerical feature matching (Section 4.4.2), and a credit score algorithm ensures the integrity of its knowledge sources (Section 4.4.3). These measures collectively maximize RAG reliability and reduce the likelihood of hallucination.

In conclusion, the error sources within LQB-IDS are well-defined and largely controlled. The dominant error source is a short-lived detection delay, while the minor LLM-related errors are effectively suppressed by our robust design. The fact that LQB-IDS still achieves superior overall accuracy compared to other state-of-the-art IoT IDSs strongly validates its practical reliability for deployment in real-world IoT environments.

6. Conclusion

In this study, an adaptive IoT IDS named LQB-IDS was proposed to enable continuous detection and self-supervised adaptation to concept drift and unknown attacks, with all processes secured by an LLM-maintained post-quantum secure blockchain. By integrating an LLM-driven analyzer for automated data labeling, an LLM-maintained post-quantum secure blockchain for robust data integrity and resistance against advanced attacks, and a knowledge distillation-based dual-switch learning mechanism for low latency and swift adaptation, the system achieved high detection accuracy, rapid responsiveness, and enhanced security. Experimental results on a hybrid IoT intrusion dataset with concept drifts and various attacks demonstrated that LQB-IDS significantly outperformed existing dynamic ML-, DL-, and LLM-based dynamic IoT IDSs in overall detection performance, adaptation speed, detection delay, and resilience against collusion attacks.

Furthermore, the deployment experiment on the computationally resource-limited IoT edge device also proves that LQB-IDS is suitable for scalability and practical deployment in real-world scenarios. This advantage is due to the system's deliberate use of a hybrid model to manage the trade-off between computational overhead and latency. While computationally intensive LLM tasks are offloaded to powerful servers, the edge detector remains lightweight.

Despite its strengths, LQB-IDS has limitations. The LLM components, while powerful, introduce computational overhead, model hallucination, and additional operational costs, as well as possible data privacy issues when using external commercial APIs. While the system incorporates mechanisms to mitigate risks like model hallucination, such as enforcing strict output formats, these intrinsic weaknesses of LLMs cannot be entirely eliminated. Additionally, the system currently relies on empirically determined thresholds for its credit scoring and collusion detection mechanisms.

Future work will focus on addressing these limitations. A primary direction is to create or fine-tune lightweight, domain-specific LLMs from open-source foundations to reduce computational overhead and enhance efficiency for IoT intrusion detection. Furthermore, we plan to develop adaptive credit scoring and collusion detection mechanisms that can dynamically learn and adjust their own thresholds, thereby improving the system's autonomy and robustness in large-scale, diverse IoT deployments.

CRedit authorship contribution statement

Yunfan Huang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft; **Maode Ma:** Conceptualization, Supervision, Writing – review & editing; **Wong Jee Keen Raymond:** Supervision, Writing – review & editing; **Chee-Onn Chow:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT, Google Gemini, and QuillBot in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Algorithms

Algorithm 3 .

Algorithm 3 DCAE Model Training Procedure.

Input:

- D_{train} : The training dataset.
- D_{val} : The validation dataset.
- α : Learning rate.
- B : Batch size.
- E_{max} : Maximum number of epochs.
- P : Early stopping patience.

Output:

- M_{best} : The state dictionary of the best-performing DCAE model.

// Initialization

```

1: Initialize DCAE model  $M$  with random weights.
2:  $optimizer \leftarrow Adam(M.parameters(), lr = \alpha)$ 
3:  $criterion \leftarrow MeanSquaredError()$ 
4:  $loss_{val\_best} \leftarrow \infty$ 
5:  $p_{counter} \leftarrow 0$ 
6:  $M_{best} \leftarrow None$ 

7: for all epoch from 1 to  $E_{max}$  do
8:    $M.train()$  ▷ Set model to training mode
9:    $loss_{train\_epoch} \leftarrow 0$ 
10:  for all each batch  $X_{batch}$  in  $D_{train}$  do
11:     $optimizer.zero_grad()$  ▷ Clear previous gradients
12:     $X'_{batch} \leftarrow M(X_{batch})$  ▷ Forward pass to get reconstructed batch
13:     $loss \leftarrow criterion(X'_{batch}, X_{batch})$  ▷ Compute reconstruction loss
14:     $loss.backward()$  ▷ Backward pass to compute gradients
15:     $optimizer.step()$  ▷ Update model weights
16:     $loss_{train\_epoch} \leftarrow loss_{train\_epoch} + loss.item()$ 
17:  end for

18:  $M.eval()$  ▷ Set model to evaluation mode
19:  $loss_{val\_epoch} \leftarrow 0$ 
20: with no_grad do
21:  for all each batch  $X_{val\_batch}$  in  $D_{val}$  do
22:     $X'_{val\_batch} \leftarrow M(X_{val\_batch})$ 
23:     $loss_{val} \leftarrow criterion(X'_{val\_batch}, X_{val\_batch})$ 
24:     $loss_{val\_epoch} \leftarrow loss_{val\_epoch} + loss_{val}.item()$ 
25:  end for
26: end with

// Early Stopping Check
27: if  $loss_{val\_epoch} < loss_{val\_best}$  then
28:    $loss_{val\_best} \leftarrow loss_{val\_epoch}$ 
29:    $p_{counter} \leftarrow 0$ 
30:    $M_{best} \leftarrow M.state\_dict()$  ▷ Save the best model state
31: else
32:    $p_{counter} \leftarrow p_{counter} + 1$ 

```

```

33:   end if
34:   if  $p_{counter} \geq P$  then
35:     Print("Early stopping triggered at epoch " + epoch)
36:     break                                ▷ Exit training loop
37:   end if
38: end for

39: return  $M_{best}$ 

```

Appendix B. Details of denoising convolutional autoencoder (DCAE) model

The DCAE is an enhanced version of the traditional autoencoder (AE) [62], incorporating two primary improvements. Firstly, it uses Triangle Area Maps (TAMs) [63] to transform the 1D input data into a 2D representation. This transformation allows for the application of convolutional layers [64] to extract contextual features, thereby enhancing the model with more powerful feature extraction capabilities. Secondly, by introducing small Gaussian noise during the training process [65], the DCAE achieves greater robustness in its detection performance. Once the DCAE flags a data sample as 'unknown,' this information, along with the data itself, is also uploaded to the blockchain to be scrutinized by the central LLM-based analyzer. Fig. B.17 shows the architecture of the DCAE model.

Appendix C. LLM prompt templates

Below is the template used by the LLM-based summarizer:

Attack Description:

<Attack Description>

(Note: This placeholder will be replaced with a detailed description of the specific attack.)

You are an IoT network security expert. Based on the description above, analyze the attack and infer the statistical characteristics of each [FEATURE] from the traffic in the following format:

“The average of [FEATURE] is [LEVEL], and its standard deviation is [LEVEL].”

Where [FEATURE] includes the following fields:
<Feature List of the Dataset>

[LEVEL] is one of the following values: [“Very Low”, “Low”, “Middle”, “High”, “Very High”]

Please strictly adhere to the format specified above

Below is the template used by the LLM-based analyzer:

“The average of [FEATURE] is [LEVEL] and its standard deviation is [LEVEL].”

In this template, [FEATURE] corresponds to each numerical feature present in the dataset, and [LEVEL] is assigned one of the following qualitative values: “Very Low”, “Low”, “Middle”, “High”, or “Very High”. The assignment of [LEVEL] is based on the percentage difference between the cluster’s average for a feature and the overall dataset’s average for that same feature, calculated as follows:

$$\text{Difference (\%)} = \frac{\text{Group Value} - \text{Dataset Value}}{\text{Dataset Value}} \times 100 \quad (\text{C.1})$$

The descriptive levels are then matched based on the following empirically derived thresholds, which are determined by multiple experiments using grid search:

- **Very Low:** Difference (%) ≤ −30 %
- **Low:** −30 % < Difference (%) ≤ −10 %
- **Middle:** −10 % < Difference (%) < 10 %
- **High:** 10 % ≤ Difference (%) < 30 %
- **Very High:** Difference (%) ≥ 30 %

This process generates a natural language description for the characteristics of each cluster.

Appendix D. Mathematical definition of evaluation metrics

$$\text{DR} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (\text{D.1})$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (\text{D.2})$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (\text{D.3})$$

$$\text{F1_score} = \frac{2 \times (\text{Precision} + \text{Recall})}{\text{Precision} + \text{Recall}} \quad (\text{D.4})$$

- **True Positive (TP):** The cases where the model correctly predicts the positive class.

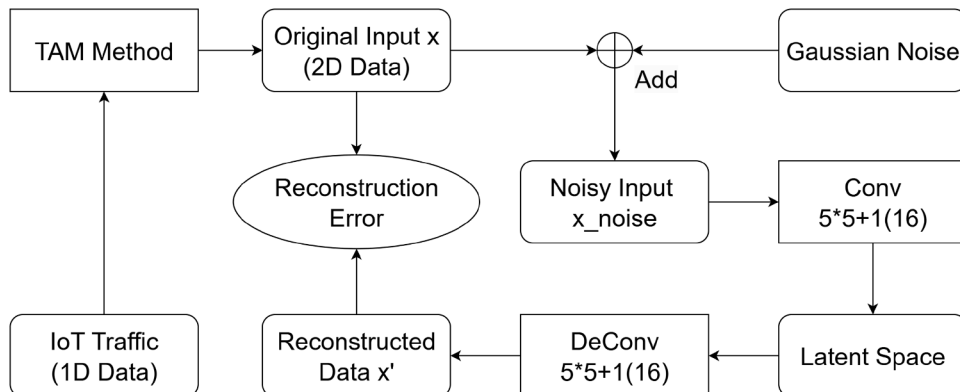


Fig. B.17. Architecture of the DCAE model.

- **True Negative (TN):** The cases where the model correctly predicts the negative class.
- **False Positive (FP):** The cases where the model incorrectly predicts the positive class when it should be negative.
- **False Negative (FN):** The cases where the model incorrectly predicts the negative class when it should be positive.

Appendix E. Glossary

Collusion Attack A coordinated attack where multiple compromised nodes send consistent malicious data [592]. LQB-IDS detects this via a credit score algorithm that identifies unusually high consistency among low-credit nodes [513, 593].

Concept Drift A phenomenon where the statistical properties of network traffic change over time, requiring the IDS to adapt [84, 161]. LQB-IDS uses a KSWIN-based detector to trigger its adaptive update mechanism [568].

DCAE (Denoising Convolutional Autoencoder) An enhanced autoencoder used in LQB-IDS to identify unknown IoT traffic [346, 384]. It transforms 1D data into a 2D representation and introduces noise during training for robustness [386, 388].

Dual-Switch Learning A novel mechanism proposed in this paper that alternates between a lightweight ML model for fast inference and an LLM-based analyzer during model updates [16, 273]. This strategy maintains high detection rates while minimizing latency [552, 553].

Knowledge Distillation A technique where a lightweight “student” model is trained by a more complex “teacher” model [582]. In LQB-IDS, the lightweight edge model is retrained using labeled data generated by the LLM-based analysis module [171, 276, 342].

LQB-IDS (LLM and Post-Quantum-Secure Blockchain-based Intrusion Detection System) The adaptive IoT IDS proposed in this paper, which synergizes an LLM, a post-quantum secure blockchain, and a dual-switch learning mechanism [9, 44, 168].

PoA (Proof-of-Authority) A lightweight blockchain consensus mechanism where trusted authority nodes validate transactions [213]. It is adopted in LQB-IDS to accommodate the computational limitations of IoT devices [213].

PQC (Post-Quantum Cryptography) Cryptographic algorithms, such as the Dilithium-based scheme used in this work, designed to be secure against attacks from both classical and future quantum computers [182, 184].

QBC (Post-Quantum-Secure Blockchain) A blockchain enhanced with Post-Quantum Cryptography (PQC) to ensure the long-term security and integrity of the IDS’s critical data against quantum threats [181, 194].

References

- [1] J. Ali, S. Kumar Singh, W. Jiang, A.M. Alenezi, M. Islam, Y. Ibrahim Daradkeh, A. Mehmood, A deep dive into cybersecurity solutions for AI-driven IoT-enabled smart cities in advanced communication networks, *Comput. Commun.* 229 (2025) 108000. <https://doi.org/10.1016/j.comcom.2024.108000>
- [2] K.O. Chee, M. Ge, G. Bai, D.D. Kim, Unveiling the evolution of IoT threats: trends, tactics, and simulation analysis, *Comput. Secur.* 157 (2025) 104537. <https://doi.org/10.1016/j.cose.2025.104537>
- [3] P. Sun, Y. Wan, Z. Wu, Z. Fang, Q. Li, A survey on privacy and security issues in IoT-based environments: technologies, protection measures and future directions, *Comput. Secur.* 148 (2025) 104097. <https://doi.org/10.1016/j.cose.2024.104097>
- [4] P. Sun, S. Shen, Y. Wan, Z. Wu, Z. Fang, X.-Z. Gao, A survey of IoT privacy security: architecture, technology, challenges, and trends, *IEEE Internet Things J.* 11 (21) (2024) 34567–34591. <https://doi.org/10.1109/JIOT.2024.3372518>
- [5] M.M. Rahman, S.A. Shakil, M.R. Mustakim, A survey on intrusion detection system in IoT networks, *Cyber Secur. Appl.* 3 (2025) 100082. <https://doi.org/10.1016/j.csa.2024.100082>
- [6] B. Eduard, *AI Tools for Protecting and Preventing Sophisticated Cyber Attacks*, IGI Global, 2023.
- [7] E. Nwafor, U. Baskota, M.S. Parwez, J. Blackstone, H. Olufowobi, Evaluating large language models for enhanced intrusion detection in internet of things networks, in: *GLOBECOM 2024–2024 IEEE Global Communications Conference*, 2024, pp. 3358–3363. ISSN: 2576–6813, <https://doi.org/10.1109/GLOBECOM52923.2024.10901300>
- [8] H. Kheddar, Transformers and large language models for efficient intrusion detection systems: a comprehensive survey, *Inf. Fusion* 124 (2025) 103347. <https://doi.org/10.1016/j.inffus.2025.103347>
- [9] Y. Gong, B.-J. Hu, A quantum-resistant key management scheme using blockchain in c-V2X, *IEEE Trans. Intell. Transp. Syst.* 25 (11) (2024) 16831–16844. <https://doi.org/10.1109/TITS.2024.3421381>
- [10] H. Gharavi, J. Granjal, E. Monteiro, Post-quantum blockchain security for the internet of things: survey and research directions, *IEEE Commun. Surv. Tutor.* 26 (3) (2024) 1748–1774. <https://doi.org/10.1109/COMST.2024.3355222>
- [11] G. Ali, A. Samuel, S.P. Kabiito, Z. Morish, A. Thomas, W. Robert, A. Denis, M. Sallam, M.M. Mijwil, J. Ayad, A.O. Salau, K. Dhoska, Integration of artificial intelligence, blockchain, and quantum cryptography for securing the industrial internet of things (IIoT): recent advancements and future trends, *Appl. Data Sci. Anal.* 2025 (2025) 19–82. <https://doi.org/10.58496/ADSA/2025/004>
- [12] U. Zukaib, X. Cui, C. Zheng, M. Hassan, Z. Shen, Meta-IDS: meta-learning-based smart intrusion detection system for internet of medical things (IoMT) network, *IEEE Internet Things J.* 11 (13) (2024) 23080–23095. <https://doi.org/10.1109/JIOT.2024.3387294>
- [13] H. Khosravi, M.A. Jabraeil Jamali, K. Majidzadeh, M. Masdari, A deep neural network-based multi-layer classifier ensembles for intrusion detection in fog-based internet of things environments, *Eng. Appl. Artif. Intell.* 140 (2025) 109727. <https://doi.org/10.1016/j.engappai.2024.109727>
- [14] S. Yu, X. Wang, Y. Shen, G. Wu, S. Yu, S. Shen, Novel intrusion detection strategies with optimal hyper parameters for industrial internet of things based on stochastic games and double deep Q-networks, *IEEE Internet Things J.* 11 (17) (2024) 29132–29145. <https://doi.org/10.1109/JIOT.2024.3406386>
- [15] A. Sharma, M. Singh, Dual replay memory reinforcement learning framework for minority attack detection, *Eng. Appl. Artif. Intell.* 144 (2025) 110083. <https://doi.org/10.1016/j.engappai.2025.110083>
- [16] T. Nay, Enhancing IoT security with AI-driven hybrid machine learning and neural network-based intrusion detection system, *Babylonian J. Artif. Intell.* 2024 (2024) 158–167. <https://doi.org/10.58496/BJAI/2024/017>
- [17] M.A. Nia, R.E. Atani, B. Fabian, E. Babulak, On detecting unidentified network traffic using pattern-based random walk, *Secur. Commun. Netw.* (2016). <https://doi.org/10.1002/sec.1557>
- [18] Z. Yin, H. Chen, H. Ma, T. Hu, L. Bai, CAEAID: an incremental contrast learning-based intrusion detection framework for IoT networks, *Comput. Netw.* 262 (2025) 111161. <https://doi.org/10.1016/j.comnet.2025.111161>
- [19] Z. Jin, J. Zhou, B. Li, X. Wu, C. Duan, FI-IIDS: a novel federated learning-based incremental intrusion detection system, *Future Gener. Comput. Syst.* 151 (2024) 57–70. <https://doi.org/10.1016/j.future.2023.09.019>
- [20] J. Mao, Z. Wei, B. Li, R. Zhang, L. Song, Toward ever-evolution network threats: a hierarchical federated class-incremental learning approach for network intrusion detection in IIoT, *IEEE Internet Things J.* 11 (18) (2024) 29864–29877. <https://doi.org/10.1109/JIOT.2024.3408634>
- [21] A. Ali, A. AlShuaibi, M.W. Arshad, An integrated federated learning framework with optimization for industrial IoT intrusion detection, *SHIFRA 2025* (2025) 110–117. <https://doi.org/10.70470/SHIFRA/2025/007>
- [22] Q. Lu, K. An, J. Li, J. Wang, Network intrusion detection for modern smart grids based on adaptive online incremental learning, *IEEE Trans. Smart Grid* 16 (3) (2025) 2541–2553. <https://doi.org/10.1109/TSG.2025.3535949>
- [23] Y. Huang, M. Ma, ILL-IDS: an incremental lifetime learning IDS for VANETs, *Comput. Secur.* 124 (2023) 102992. <https://doi.org/10.1016/j.cose.2022.102992>
- [24] Y. Huang, M. Ma, AILL-IDS: an automatic incremental lifetime learning intrusion detection system for vehicular ad hoc networks, *IEEE Trans. Intell. Transp. Syst.* 26 (2) (2025) 2669–2678. <https://doi.org/10.1109/TITS.2024.3510584>
- [25] M.N. Halgamuge, D. Niyato, Adaptive edge security framework for dynamic IoT security policies in diverse environments, *Comput. Secur.* 148 (2025) 104128. <https://doi.org/10.1016/j.cose.2024.104128>
- [26] Z. Chen, H. Zou, T. Hu, X. Yuan, X. Fang, Y. Pan, J. Li, HC-NIDS: historical contextual information based network intrusion detection system in internet of things, *Comput. Secur.* 152 (2025) 104367. <https://doi.org/10.1016/j.cose.2025.104367>
- [27] A. Gueriani, H. Kheddar, A.C. Mazari, M.C. Ghanem, A robust cross-domain IDS using biGRU-LSTM-attention for medical and industrial IoT security, *ICT Express* (2025). <https://doi.org/10.1016/j.icte.2025.08.011>
- [28] Z. Xiao, H. Xing, R. Qu, H. Li, H. Tong, S. Luo, J. Song, L. Feng, Q. Wan, Knowledge aggregation transformer network for multivariate time series classification, *IEEE Trans. Big Data* (2025) 1–17. <https://doi.org/10.1109/TBDATA.2025.3594294>
- [29] Z. Xiao, H. Xing, R. Qu, H. Li, X. Cheng, L. Xu, L. Feng, Q. Wan, Heterogeneous mutual knowledge distillation for wearable human activity recognition, *IEEE Trans. Neural Netw. Learn. Syst.* 36 (9) (2025) 16589–16603. <https://doi.org/10.1109/TNNLS.2025.3556317>
- [30] M.A. Ferrag, M. Ndhlovu, N. Tihanyi, L.C. Cordeiro, M. Debbah, T. Lestable, N.S. Thandi, Revolutionizing cyber threat detection with large language models: a privacy-preserving BERT-based lightweight model for IIoT devices, *IEEE Access* 12 (2024) 23733–23750. <https://doi.org/10.1109/ACCESS.2024.3363469>

- [31] T. Ali, P. Kostakos, HuntGPT: Integrating Machine Learning-Based Anomaly Detection and Explainable AI with Large Language Models (LLMs), 2023. <https://doi.org/10.48550/arXiv.2309.16021>
- [32] Y. Li, Z. Xiang, N.D. Bastian, D. Song, B. Li, IDS-agent: an LLM agent for explainable intrusion detection in IoT networks, 2024.
- [33] X. Hu, H. Chen, H. Bao, W. Wang, F. Liu, G. Zhou, P. Yin, A LLM-based agent for the automatic generation and generalization of IDS rules, in: 2024 IEEE 23rd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2024, pp. 1875–1880. ISSN: 2324–9013, <https://doi.org/10.1109/TrustCom63139.2024.00259>
- [34] B. Xiao, B. Kantarci, J. Kang, D. Niyato, M. Guizani, Efficient prompting for LLM-based generative internet of things, IEEE Internet Things J. 12 (1) (2025) 778–791. <https://doi.org/10.1109/JIOT.2024.3470210>
- [35] M.S. Sani, S. Iranmanesh, H. Salarian, R. Raad, A. Jamalipour, BIDS: blockchain-enabled intrusion detection system in smart cities, IEEE Internet Things Mag. 7 (2) (2024) 107–113. <https://doi.org/10.1109/IOTM.001.2300191>
- [36] M.T. Naz, W. Elmedany, M. Ali, Securing SCADA systems in smart grids with IoT integration: a self-defensive post-quantum blockchain architecture, Internet Things 28 (2024) 101381. <https://doi.org/10.1016/j.iot.2024.101381>
- [37] A. Diaf, A.A. Korba, N. Elislem Karabadi, Y. Ghamri-Doudane, BARTPredict: Empowering IoT security with LLM-driven cyber threat prediction, in: GLOBECOM 2024–2024 IEEE Global Communications Conference, 2024, pp. 1239–1244. ISSN: 2576–6813, <https://doi.org/10.1109/GLOBECOM52923.2024.10901770>
- [38] B.D. Son, N.T. Hoa, T.V. Chien, W. Khalid, M.A. Ferrag, W. Choi, M. Debbah, Adversarial attacks and defenses in 6G network-assisted IoT systems, IEEE Internet Things J. 11 (11) (2024) 19168–19187. <https://doi.org/10.1109/JIOT.2024.3373808>
- [39] G. Chen, X. Li, L. You, A.M. Abdelmoniem, Y. Zhang, C. Yuen, A data poisoning resistible and privacy protection federated-learning mechanism for ubiquitous IoT, IEEE Internet Things J. 12 (8) (2025) 10736–10750. <https://doi.org/10.1109/JIOT.2024.3514637>
- [40] M. Bhatia, S. Sood, Quantum-computing-inspired optimal power allocation mechanism in edge computing environment, IEEE Internet Things J. 11 (10) (2024) 17878–17885. <https://doi.org/10.1109/JIOT.2024.3358900>
- [41] D. Namakshenas, A. Yazdinejad, A. Dehghantanha, G. Srivastava, Federated quantum-based privacy-preserving threat detection model for consumer internet of things, IEEE Trans. Consum. Electron. 70 (3) (2024) 5829–5838. <https://doi.org/10.1109/TCE.2024.3377550>
- [42] M.A. Khan, D. Puri, Challenges and opportunities in implementing quantum-safe key distribution in IoT devices, in: 2024 3rd International Conference for Innovation in Technology (INOCON), 2024, pp. 1–7. <https://doi.org/10.1109/INOCON60754.2024.10511390>
- [43] R. Xu, S.R. Pokhrel, Q. Lan, G. Li, Post Quantum Secure Blockchain-based Federated Learning for Mobile Edge Computing, 2023. arXiv:2302.13258 [cs], <https://doi.org/10.48550/arXiv.2302.13258>
- [44] G. Al-Kateb, GAM: introducing a novel algorithm for advanced ultralightweight cryptography for IoT devices, Babylonian J. Internet Things 2025 (2025) 131–142. <https://doi.org/10.58496/BJIoT/2025/008>
- [45] A.G. Kumar, A. Rastogi, V. Ranga, Evaluation of different machine learning classifiers on new IoT dataset CICIoT2023, in: 2024 International Conference on Intelligent Systems for Cybersecurity (ISCS), 2024, pp. 1–6. <https://doi.org/10.1109/ISCS61804.2024.10581375>
- [46] Y. Jung, T. Kim, M.-R. Han, S. Kim, G. Kim, S. Lee, Y.J. Choi, Ovarian tumor diagnosis using deep convolutional neural networks and a denoising convolutional autoencoder, Sci. Rep. 12 (1) (2022) 17024. <https://doi.org/10.1038/s41598-022-20653-2>
- [47] J. Villegas, S. Fortes, J. Cantizani-Esteva, J.R. Suarez, R.M. Cuervo, R. Barco, A scoring framework for the unsupervised identification of the relevance of metrics in cell degradation clusters, IEEE Commun. Lett. 28 (10) (2024) 2288–2292. <https://doi.org/10.1109/LCOMM.2024.3411768>
- [48] G. Quadrio, R. Pompa, E. Bassetti, C. D'elia, G. Scacco, M. Conti, “They will adapt”: improving anomaly detection on IoT networks through continuous learning, in: 2024 20th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob), 2024, pp. 1–6. ISSN: 2160–4894, <https://doi.org/10.1109/WiMob61911.2024.10770308>
- [49] Z. Zhang, H. Zhang, An online transfer learning model for intrusion detection using FT-transformer and KSWIN-driven concept drift detection mechanism, in: 2024 5th International Conference on Computer Engineering and Application (ICCEA), 2024, pp. 128–131. ISSN: 2159–1288, <https://doi.org/10.1109/ICCEA62105.2024.10603489>
- [50] Z. Wang, R. Zhou, S. Yang, D. He, S. Chan, A novel lightweight IoT intrusion detection model based on self-knowledge distillation, IEEE Internet Things J. 12 (11) (2025) 16912–16930. <https://doi.org/10.1109/JIOT.2025.3533092>
- [51] M.A. Burhanuddin, Secure and scalable quantum cryptographic algorithms for next-generation computer networks, KHARIZMIA 2023 (2023) 95–102. <https://doi.org/10.70470/KHARIZMIA/2023/009>
- [52] K. Ramesh, N.C. Miller, A. Faridi, F. Aloul, I. Zulkernan, A.R. Sajun, Efficient machine learning frameworks for strengthening cybersecurity in internet of medical things (IoMT) ecosystems, in: 2024 IEEE International Conference on Internet of Things and Intelligence Systems (IoTIS), 2024, pp. 92–98. ISSN: 2832–1383, <https://doi.org/10.1109/IoTIS64014.2024.10799438>
- [53] G.P. Nilesh, V.P. Dipak, Binary and multiclass classification intrusion detection system using benchmark NSL-KDD and machine learning models, in: 2024 International Conference on Data Science and Network Security (ICDSNS), 2024, pp. 1–7. <https://doi.org/10.1109/ICDSNS62112.2024.10691256>
- [54] S. Vellingiri, N.G.B. Amma, N.-K. Baik, Detection of DoS attacks in smart city networks with feature distance maps: a statistical approach, IEEE Internet Things J. 10 (21) (2023) 18853–18860. <https://doi.org/10.1109/JIOT.2023.3264670>
- [55] R. Xu, S.R. Pokhrel, Q. Lan, G. Li, FAIR-BFL: flexible and incentive redesign for blockchain-based federated learning, in: Proceedings of the 51st International Conference on Parallel Processing, ICPP '22, Association for Computing Machinery, New York, NY, USA, 2023. <https://doi.org/10.1145/3545008.3545040>
- [56] R. Xu, WhiteByeBye/Post-Quantum-Secure-Blockchained-Federated-Learning, 2025. original-date: 2023-09-11T13:00:58Z, <https://github.com/WhiteByeBye/Post-Quantum-Secure-Blockchained-Federated-Learning>
- [57] K.A. Delgado-Vargas, C.M. Lopez, G. Gallegos-Garcia, Implementation of post-quantum cryptographic libraries on ARM architectures, in: 2024 IEEE Biennial Congress of Argentina (ARGENCON), 2024, pp. 1–4. <https://doi.org/10.1109/ARGENCON62399.2024.10735905>
- [58] O. Team, GPT-4o System Card, 2024. arXiv:2410.21276 [cs], <https://doi.org/10.48550/arXiv.2410.21276>
- [59] L. Team, Llama-Nemotron: Efficient Reasoning Models, 2025. arXiv:2505.00949 [cs], <https://doi.org/10.48550/arXiv.2505.00949>
- [60] G. Team, Gemini: A Family of Highly Capable Multimodal Models, 2025. arXiv:2312.11805 [cs], <https://doi.org/10.48550/arXiv.2312.11805>
- [61] D. Team, DeepSeek-V3 Technical Report, 2025. arXiv:2412.19437 [cs], <https://doi.org/10.48550/arXiv.2412.19437>
- [62] P.V. Dinh, D.T. Hoang, N.Q. Uy, D.N. Nguyen, S.P. Bao, E. Dutkiewicz, Multiple-input auto-encoder for IoT intrusion detection systems with heterogeneous data, in: ICC 2024 - IEEE International Conference on Communications, 2024, pp. 2707–2712. ISSN: 1938–1883, <https://doi.org/10.1109/ICC51166.2024.10622942>
- [63] Y. Zhou, S. Gao, D. Guo, X. Wei, J. Rokne, H. Wang, A survey of change point detection in dynamic graphs, IEEE Trans. Knowl. Data Eng. 37 (3) (2025) 1030–1048. <https://doi.org/10.1109/TKDE.2024.3523857>
- [64] G.-Q. Zeng, Y.-W. Yang, K.-D. Lu, G.-G. Geng, J. Weng, Evolutionary adversarial autoencoder for unsupervised anomaly detection of industrial internet of things, IEEE Trans. Reliab. (2025) 1–15. <https://doi.org/10.1109/TR.2025.3528256>
- [65] F.S. Alrayes, M. Zakariah, S.U. Amin, Z. Iqbal Khan, M. Helal, Intrusion detection in IoT systems using denoising autoencoder, IEEE Access 12 (2024) 122401–122425. <https://doi.org/10.1109/ACCESS.2024.3451726>